

# Cross-city path-loss transfer for digital-twin network planning: why tile parameters fall short and how site-aware link models close the gap

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## Abstract

Digital-twin network planning needs path-loss models for areas without measurements. Tile-based methods fit a log-distance exponent and intercept per map tile and transfer them to unmeasured tiles by descriptor similarity. We study this transfer across cities on ray-traced corpora: 11 geo-referenced cities with 220 rooftop sites at 3.5 GHz (8,977,400 links) and 7.5 GHz, a random-siting replicate and independently modelled external scenes. The tile labels are imprecise: their cluster-robust uncertainty is 3.2 times the naive one, and they change with the serving sites and the dynamic range of the data. Learned regressors transfer them better than the source median in most cities, though not significantly, yet even perfect tile labels give a path-loss error of 11.6 dB and the correct serving site for only 50.1% of the pixels, and composing the parameters of the tiles a link crosses helps with exact labels but not once they are transferred. A site-aware hurdle model that predicts each link, and whether it exists, from its obstruction and diffraction geometry reduces the error on unseen cities to 9.3 dB, raises serving-site accuracy to 68.8% and cuts the coverage lost when planning five sites from 20.6% to 0.5%, in every held-out city. On external scenes with other materials, a survey of 25 random tiles calibrates it to the accuracy of exact tile labels. A city-level certificate lets it promise 92.0% of the pixels as covered at a guaranteed risk, against 78.3% for tile transfer. All results are reproducible by one command.

**Keywords:** digital twin, network planning, path-loss modelling, transfer learning, ray tracing, coverage certification, empirical Bayes

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## 1. Introduction

Network planning with a digital twin requires a propagation model for places where no measurement has been taken yet: coverage, interference and site selection must be assessed before a campaign exists in the target region [1, 2]. Ray tracing resolves the geometry but must be rerun for every change of layout and needs detailed 3-D data [3, 4]; empirical formulas such as Okumura–Hata or 3GPP TR 38.901 are cheap but apply one parameter set to a whole environment class [5, 6]. Tile-based models sit between the two: the twin is partitioned into tiles, each tile carries the two parameters  $(\gamma, P_{L_0})$  of a log-distance law, and parameters learned where measurements exist are transferred to unmeasured tiles by environmental similarity [7, 8]. The transported quantity is small and physically interpretable, and a planning tool can evaluate it for any site layout.

Whether such transfer works *across cities*, and what its errors cost a network planner, has not been established. The

descriptor-based framework of [8], which transfers tile parameters with a Gaussian kernel over a descriptor space, was evaluated on confidential measurements from two cities, one serving as the source of the other. Three questions remain open. First, how reliable are the tile labels themselves, given that a tile of 100 m sees its serving sites over a narrow range of distances and that neighbouring receivers share propagation paths? Second, how does kernel transfer behave when the target city differs from every source city, which is exactly when transfer is needed, and how does it compare with learned regressors under one protocol? Third, what do parameter errors mean for the quantities a planner uses: path loss, serving cell, SINR, coverage, cell edges and the choice of sites, is the tile parameter the right quantity to transfer at all, and how much coverage can an operator promise from a transferred model?

This paper answers these questions on a public benchmark. Our contributions are the following.

1. **Reproducible multi-city, multi-band corpora.** Eleven geo-referenced city scenes built from a pinned release of open building data are ray traced from 220 rooftop sites at 3.5 GHz (8,977,400 links, 10,428 labelled tiles) and at 7.5 GHz in the upper mid-band, with a ray-budget study, a physical consistency check of the solver output, a replicate with random rooftop siting and an external test on inde-

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pends on the geometry of the scene. Every number in this paper is regenerated from the released data by one command.

2. **Tile labels with honest precision.** We show that the per-tile fit is ill-conditioned (Proposition 3), replace ad-hoc regularisation by empirical-Bayes shrinkage whose risk is bounded whatever the tile geometry (Proposition 4) with a small-sample (CR2) cluster-robust variance, and show that the labels depend on the serving sites, the tile size and the dynamic range of the data.
3. **An analysis of kernel transfer.** We prove that the Gaussian kernel operator has two limiting regimes (Theorem 1), measure where the tuned operator lies on every held-out city, and show that learning its metric removes most of its disadvantage.
4. **A leave-one-city-out benchmark with city-level inference.** Standard-model, label-averaging, Gaussian-process, linear, tree, neural and domain-adaptation operators are compared under one tuning protocol with five seeds, with and without surveyed tiles (random, contiguous or designed), and every claim is tested at the level of cities with bootstrap intervals, Wilcoxon tests and Holm correction.
5. **Network-level evaluation and site-aware link transfer.** We derive what a tile model can express for cell association and SINR (Proposition 1), evaluate every operator through path loss, serving cell, SINR, coverage, cell edges, dimensioning and site planning scored on the ray-traced truth, and propose a hurdle link model conditioned on the obstruction geometry of each link that outperforms even the tile model with perfect labels, in receiver-tile and in mixed-path composition, in every held-out city of the corpus, and test it across bands, sitings and independently modelled scenes.
6. **Survey design and coverage certificates.** We give a survey design for MDT-style site calibration with a  $(1 - 1/e)$  guarantee (Proposition 5) and a distribution-free, city-level coverage certificate from leave-one-city-out residuals (Proposition 6), and measure what each transfer model can promise at a stated risk.

Section 2 reviews related work, Section 3 formalises the network model and the transfer problem, Section 4 describes the corpus, labels, descriptors, operators and evaluation, Section 5 reports the results, Section 6 discusses their implications and limitations, and Section 7 concludes. Proofs are in Appendix A, further robustness tables in Appendix C and the complete city-level comparisons in Appendix D.

## 2. Related work

*Empirical and site-specific propagation models.* Empirical models such as Okumura–Hata [5] and the 3GPP TR 38.901 formulas [6] express path loss through a small number of parameters fitted to measurement campaigns and applied uniformly over an environment class. Measurement studies showed early that the path-loss exponent is itself a random quantity that changes from cell to cell [9], which is the variability a tile-level model tries to capture. Deterministic ray tracing

resolves the geometry explicitly [3]; differentiable ray tracers such as Sionna RT make city-scale simulation and calibration against measurements practical [4, 10]. We use ray tracing to produce ground truth for many cities, not as the run-time predictor.

*Learned radio maps and path-loss regression.* Convolutional [11], autoencoder [12], adversarial [13] and diffusion [14] models predict dense radio maps from environment maps, and radio-map estimation from sparse measurements has a mature statistical treatment [15]. Channel knowledge maps [16] generalise the idea to other channel quantities. Regression of path loss from engineered or image features has been studied with measurements at 2.6 GHz [17] and 28 GHz [18], across several cities with obstruction-height maps [19], and is surveyed in [20]. These methods predict path loss or maps directly; the tile-parameter approach studied here predicts a compact, physically interpretable pair per tile that a planning tool can evaluate for any site layout.

*Tile-based transfer and digital twins.* Digital twins of wireless networks [1, 2] need propagation models for regions where no measurements exist yet. MAPLE divides 3-D city maps into tiles and estimates mixed-path propagation in a new deployment area from the similarity of its tiles to tiles of measured areas [7]. The descriptor-based framework of [8] describes each tile by 116 terrain, built-structure, land-cover and climatic descriptors, calibrates log-distance parameters per tile from measurements with the mixed-path composition of MAPLE, and transfers them to unmeasured tiles by a Gaussian-kernel weighting of descriptor similarity. It was evaluated with confidential measurements of an industrial partner in two European cities, each serving once as the source of the other, with tiles of 200 m to 1000 m, by the normalised error of the received power, and against empirical models only. We study the same transfer problem on a larger public corpus, analyse the kernel operator, compare it with learned regressors under a common protocol, evaluate both the receiver-tile and the mixed-path composition, and measure the consequences for network planning.

*Transfer under covariate shift and planning.* Transfer between domains with different input distributions is covariate shift [21], and its difficulty depends on the discrepancy between the domains [22]; importance weighting [21] and the alignment of feature statistics [23] are standard remedies, which we include as baselines. In operational networks, targeted measurements can be collected with the minimisation-of-drive-tests (MDT) framework [24, 25], which motivates our few-shot setting; choosing where to measure is a submodular sensor-placement problem [26]. Site selection with propagation predictions is a classical optimisation problem [27]; coverage maximisation is submodular and admits greedy approximation guarantees [28].

*Uncertainty and censored measurements.* Conformal prediction gives distribution-free prediction sets under exchangeability [29], and leave-one-out variants such as the jackknife+ reuse

every training point for calibration [30]; we apply the exchangeability argument at the level of cities, the unit of shift. Measurements with a detection floor are censored, for which the Tobit model is the classical treatment [31]; received-power data beyond a receiver's sensitivity are censored in exactly this sense.

### 3. System model and problem formulation

#### 3.1. Network and propagation model

A city contains a set  $\mathcal{B}$  of base-station sites and a set  $\mathcal{P}$  of receiver locations (pixels) at user height. The link  $(p, b)$  has three-dimensional length  $d_{pb}$  and path loss  $P_L(p, b)$  in dB. With transmit power  $P_{tx}$  (dBm) at every site and noise power  $N$ , the received power, serving site and downlink signal-to-interference-plus-noise ratio under full load are

$$P_{pb} = P_{tx} - P_L(p, b), \quad s(p) = \arg \max_{b \in \mathcal{B}} P_{pb},$$

$$\text{SINR}(p) = \frac{10^{P_{p, s(p)}/10}}{\sum_{b \neq s(p)} 10^{P_{pb}/10} + 10^{N/10}}. \quad (1)$$

A pixel is *covered* at coupling-loss level  $L$  if  $P_L(p, s(p)) \leq L$ , and its spectral efficiency is obtained from the SINR with the attenuated and truncated Shannon mapping used in 3GPP system evaluations [32].

The city is partitioned into square tiles of side  $G_s$ . Inside tile  $t$  the path loss follows the log-distance law [33, 9]

$$P_L(p, b) = P_{L_{0,t}} + 10 \gamma_t \log_{10} \frac{d_{pb}}{d_0} + \varepsilon_{pb}, \quad p \in t, \quad (2)$$

with exponent  $\gamma_t$ , intercept  $P_{L_{0,t}}$  at the reference distance  $d_0$  and a residual  $\varepsilon_{pb}$  of variance  $\sigma_\varepsilon^2$ . We use  $d_0 = 100$  m, the tile scale, so that  $P_{L_{0,t}}$  is an interpolated rather than an extrapolated quantity; the intercept at 1 m follows as  $P_{L_{0,t}} - 20\gamma_t$ . Writing  $x = 10 \log_{10}(d/d_0)$ , Eq. (2) is linear in  $\theta_t = (\gamma_t, P_{L_{0,t}})^\top$  with design matrix  $\mathbf{A}_t = [\mathbf{x} \ \mathbf{1}]$ .

Equation (2) is the interface between the digital twin and network planning: a planning tool that knows  $\theta_t$  for every tile and the site positions can evaluate Eq. (1) anywhere without ray tracing. Two structural consequences of the tile model are used later.

**Proposition 1** (What a receiver-tile model can express). *Let every link seen from pixel  $p$  use the parameters of the tile containing  $p$ , with  $\gamma_t > 0$ . Then (i) the predicted serving site is the nearest site,  $s(p) = \arg \min_b d_{pb}$ , whatever  $\theta_t$ ; and (ii) the predicted SINR is*

$$\text{SINR}(p) = \frac{(d_{ps}/d_0)^{-\gamma_t}}{\sum_{b \neq s} (d_{pb}/d_0)^{-\gamma_t} + 10^{(N - P_{tx} + P_{L_{0,t}})/10}}, \quad (3)$$

*so in the interference-limited regime it depends on the exponent only, while coverage at level  $L$  depends on both parameters.*

The proof is immediate from Eqs. (1) and (2). Part (i) bounds what any receiver-tile predictor can achieve in cell association: the metric is identical for every transfer operator and measures

the tile model itself. Part (ii) identifies which parameter errors matter for which planning quantity. Mixed-path compositions [7, 8], which combine the parameters of every tile a link crosses, are not bound by Proposition 1 and are evaluated in Section 5.7.

#### 3.2. The transfer problem

Each tile carries a descriptor  $\mathbf{v}_t \in \mathbb{R}^p$  computed from the geometry of the digital twin. Measurements, and hence labels  $\theta_t$ , are available in a set of *source* cities  $\mathcal{T}_S = \{(\mathbf{v}_j, \theta_j)\}_{j=1}^N$  and not in the *target* city, whose tiles have descriptors only. A *transfer operator* maps the source data and a target descriptor to a parameter estimate,  $\hat{\theta}_u = \mathcal{A}(\mathcal{T}_S; \mathbf{v}_u)$ . Source and target descriptors follow different distributions: the purpose of transfer is to serve cities unlike those measured, so covariate shift [21, 22] is the defining feature of the problem. In the *few-shot* variant,  $k$  target tiles are surveyed and their labels revealed before prediction.

The operator is judged at two levels: the parameter error on the target tiles and, more importantly for planning, the error of the network quantities of Section 3.1 computed from the predicted parameters and compared with the ray-traced network.

#### 3.3. Risk of label-averaging operators

Many transfer operators, including the Gaussian-kernel operator of [8] and  $k$ -nearest-neighbour averaging, predict a convex combination of source labels,  $\hat{\theta}_u = \sum_j w_j(\mathbf{v}_u) \theta_j$  with  $w_j \geq 0$ ,  $\sum_j w_j = 1$  and weights that depend on descriptors only.

**Proposition 2** (Risk decomposition). *Let the source labels be  $\theta_j = \theta^*(\mathbf{v}_j) + \eta_j$ , with  $\theta^*(\mathbf{v})$  the conditional mean parameter and  $\eta_j$  zero-mean label noise of covariance  $\Sigma_\eta$ , independent across  $j$  and of the descriptors. For a label-averaging operator the target risk  $R = \mathbb{E}_{\mathbf{v}} \|\hat{\theta}(\mathbf{v}) - \theta^*(\mathbf{v})\|^2$  is*

$$R = \underbrace{\mathbb{E}_{\mathbf{v}} \left\| \sum_j w_j(\mathbf{v}) \theta^*(\mathbf{v}_j) - \theta^*(\mathbf{v}) \right\|^2}_{\text{smoothing bias}} + \underbrace{\text{tr}(\Sigma_\eta) \mathbb{E}_{\mathbf{v}} \sum_j w_j(\mathbf{v})^2}_{\text{noise transmission}}. \quad (4)$$

The second term equals  $\text{tr}(\Sigma_\eta) \mathbb{E}[1/n_{\text{eff}}]$  with the effective neighbourhood size  $n_{\text{eff}}(\mathbf{v}) = 1/\sum_j w_j(\mathbf{v})^2$ . Concentrated weights lower the bias and raise the noise transmission; diffuse weights do the opposite. Under covariate shift the bias term grows, because the source set may contain no tile whose conditional mean resembles the target's; with noisy labels the second term matters as well. Ridge regression is also linear in the labels, and so is a regression tree once its partition is fixed; the difference is that their weights are not confined to the probability simplex and are shaped by the descriptor-label relation rather than fixed by an isotropic distance, which is the distinction that matters below.

### 4. Methods

Figure 1 summarises the pipeline. Every stage is implemented in the accompanying repository and executed by a single command; every number reported in Sections 5–6 is generated from its outputs.

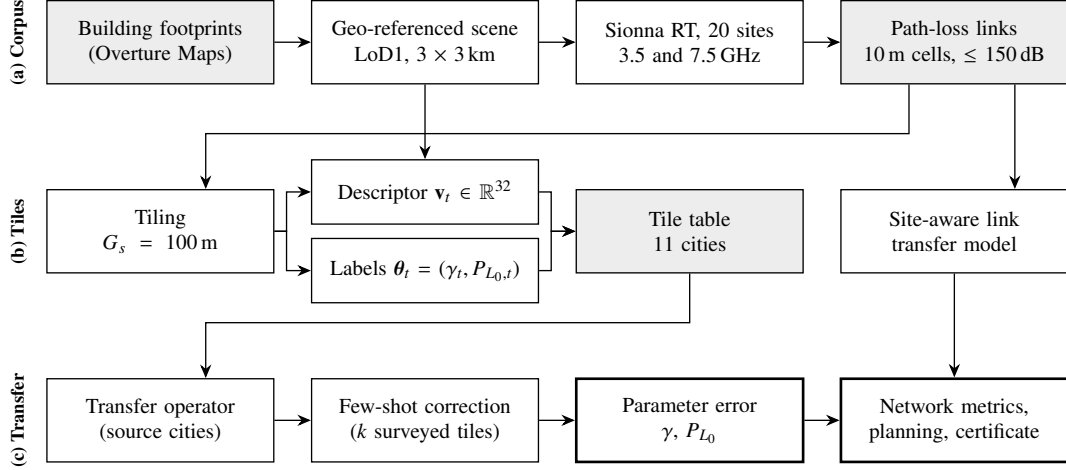


Figure 1: Processing pipeline. (a) Geo-referenced scenes are built from open building footprints and ray traced. (b) Links are grouped into tiles; each tile receives a label  $\theta_t = (\gamma_t, P_{L0,t})$  and a descriptor  $\mathbf{v}_t \in \mathbb{R}^{32}$  computed from the same scene geometry. (c) Tile transfer operators fitted on source cities predict the labels of a held-out city, optionally corrected with  $k$  surveyed tiles; the site-aware link model predicts every link from its geometry. Both are evaluated through the network quantities of Section 3.1, a site plan and a coverage certificate.

#### 4.1. Ray-traced corpus

Each city is a square of side 3 km centred on a fixed coordinate and expressed in a local metric frame, the UTM projection of its zone shifted to the centre [34]. Building footprints are taken from the buildings theme of Overture Maps [35], release 2026-09-23.0, a versioned open dataset whose primary source is OpenStreetMap [36]; a pinned release makes the geometry exactly reproducible, which a live map query is not. Because grid north and true north differ, the footprint query box covers all four corners of the square. Building heights are resolved in order from the height attribute, from the number of floors at 3.2 m per floor, and otherwise from the median of the resolved heights of the city; the share of each source is reported per city. Footprint exteriors are clipped to the square, parts smaller than 4 m<sup>2</sup> are discarded, and the remainder is extruded to level-of-detail 1 and placed on a flat ground plane. Buildings use the ITU concrete model and the ground the ITU medium-dry-ground model [37].

Each city has 20 sites on rooftops: local height maxima of the building mesh on a 25 m grid are ranked by height and selected greedily with a minimum separation of 250 m, and the antenna is mounted 6 m above the roof; if too few candidates satisfy the separation, the remaining sites are drawn at random from the other rooftop candidates with a fixed seed. Radio maps are computed with Sionna RT [4, 10] (version 2.1.0) at 3.5 GHz with isotropic antennas,  $1 \times 10^8$  rays per site, maximum interaction depth 5, specular and diffuse reflection, refraction and diffraction (including edge diffraction) enabled, on a 10 m grid at 1.5 m receiver height. Cells with path loss above 150 dB, beyond any practical coupling loss, are discarded. The solver version and settings are stored with every city. Twenty sites on 9 km<sup>2</sup> give one site per 0.45 km<sup>2</sup>, the density of a hexagonal layout with an inter-site distance of about 720 m, sparser than the 500 m of the 3GPP urban-macro scenario [6]; ranking rooftops by height follows macro-cell practice. Both choices are tested in Section 5.10.

Two further corpora use the same scenes and settings. The *second band* re-traces every site at 7.5 GHz, in the 7 GHz to 24 GHz range studied for NR [38]. The *siting replicate* re-traces 4 cities (Amsterdam, London, New York, Paris) at 3.5 GHz with the same number of sites taken from the rooftop candidates in random order instead of by height, under the same separation rule.

*Consistency check.* The planar radio-map estimator of the solver weights a ray crossing the measurement plane by the inverse cosine of its incidence angle, and a diffracted ray by an integration weight over the diffraction cone; the first weight is unbounded and the second can be very large, so a cell occasionally receives one overweighted contribution. Its path loss can then fall below free space by tens of decibels, which no incoherent sum of a few propagation paths allows. We remove every link whose path loss lies more than 6 dB below the free-space loss at its 3-D distance: 1,170 links, 0.013% of the corpus. For the site with the most such links, disabling diffraction reduces the links more than 6 dB below free space from 78 to 0 with otherwise identical settings, which identifies the diffraction weights as their source.

#### 4.2. Tile labels

Links are assigned to tiles of side  $G_s = 100$  m by their receiver position; tiles with fewer than 30 links are not labelled. Inside a tile, the ordinary least-squares (OLS) slope is poorly determined because every receiver sees the serving sites from a similar distance.

**Proposition 3** (Ill-conditioning of the per-tile fit). *For  $\mathbf{A} = [\mathbf{x} \ \mathbf{1}]$  with  $n$  rows,  $\det(\mathbf{A}^\top \mathbf{A}) = n^2 \text{Var}(x)$  and  $\text{Var}(\hat{y}_{\text{OLS}}) = \sigma_\varepsilon^2 / (n \text{Var}(x))$ , which diverges as the intra-tile spread of  $x$  vanishes.*

A receiver at distance  $d$  from its site sees a tile of side  $G_s$  span only  $\Delta x \approx 10 \log_{10}(1 + G_s/d)$ , which is 0.41 dB for  $G_s = 100$  m

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**Algorithm 1:** Tile labels

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**Input:** links  $(x_i, y_i, t_i, b_i)$  of one city; minimum count  $n_{\min}$

**Output:**  $\hat{\theta}_t, w_t$ , iid and cluster-robust standard errors

```

1 foreach tile  $t$  with  $n_t \geq n_{\min}$  do
2    $\bar{x}_t, \bar{y}_t, S_{xx,t}, S_{xy,t}; \quad b_t \leftarrow S_{xy,t}/S_{xx,t}; \quad v_t \leftarrow$ 
   cluster-robust variance of  $b_t$  (Eq. (6))
3 end
4  $(\hat{\mu}, \hat{\tau}^2) \leftarrow$  DerSimonian–Laird over  $\{(b_t, v_t)\}$ 
5 foreach tile  $t$  do
6    $w_t \leftarrow v_t/(v_t + \hat{\tau}^2); \quad \hat{\gamma}_t \leftarrow (1 - w_t)b_t + w_t\hat{\mu};$ 
    $\hat{P}_{L_0,t} \leftarrow \bar{y}_t - \hat{\gamma}_t\bar{x}_t$ 
7   residuals  $e_i \leftarrow y_i - \hat{P}_{L_0,t} - \hat{\gamma}_tx_i$ 
8   standard errors from Eq. (6) with iid and with site
   clusters
9 end

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and  $d = 1$  km. We therefore shrink the slope towards the city-wide mean with weights set by the data (empirical Bayes [39]).

**Proposition 4** (Empirical-Bayes shrinkage). *Let  $b_t$  be the OLS slope of tile  $t$ , computed with  $x$  centred inside the tile, with sampling variance  $v_t$ , and let the true slopes vary between tiles with mean  $\mu$  and variance  $\tau^2$ . Among estimators  $\hat{\gamma}_t = (1 - w)b_t + w\mu$  the mean squared error  $(1 - w)^2v_t + w^2\tau^2$  is minimised by*

$$w_t = \frac{v_t}{v_t + \tau^2} = \frac{\lambda_t}{\lambda_t + S_{xx,t}}, \quad \lambda_t = \frac{S_{xx,t} v_t}{\tau^2}, \quad (5)$$

which is ridge regression on the centred slope with penalty  $\lambda_t$  and prior mean  $\mu$ . Its risk is  $v_t\tau^2/(v_t + \tau^2) \leq \min(v_t, \tau^2)$ , bounded by  $\tau^2$  however degenerate the tile geometry.

Radio-map cells served by the same site share propagation paths, so the independent-residual variance  $s_t^2/S_{xx,t}$  understates  $v_t$ . We therefore use the cluster-robust variance of Eq. (6) (below) of the OLS slope, with the serving sites as clusters, never below the independent value; a tile served by a single site uses the independent value multiplied by the median ratio of the two over the city (its design effect). We estimate  $(\mu, \tau^2)$  per city with the DerSimonian–Laird moment estimator [40] over the tiles with a non-degenerate span, set  $\hat{\gamma}_t = (1 - w_t)b_t + w_t\hat{\mu}$ , and leave the intercept unpenalised:  $\hat{P}_{L_0,t} = \bar{y}_t - \hat{\gamma}_t\bar{x}_t$ . A tile with no span takes  $\hat{\gamma}_t = \hat{\mu}$ . The shrinkage weight  $w_t$  is published with every label. Algorithm 1 states the procedure; it is vectorised over all links of a city.

**Label precision.** The estimator is linear in the path losses. In the centred parametrisation  $y_i = a_t + \gamma_t(x_i - \bar{x}_t) + e_i$  with design rows  $\mathbf{z}_i = (1, x_i - \bar{x}_t)$ , the coefficients are  $\mathbf{M} \sum_i \mathbf{z}_i y_i$  up to the prior term, with  $\mathbf{M} = \text{diag}(1/n, 1/(S_{xx,t} + \lambda_t))$  ( $\lambda_t = 0$  for the OLS slope), and  $\hat{P}_{L_0,t} = \hat{a}_t - \hat{\gamma}_t\bar{x}_t$ . With the links of the tile partitioned by serving site into  $G$  clusters  $\mathcal{I}_1, \dots, \mathcal{I}_G$ , with design blocks  $\mathbf{Z}_c$  and residuals  $\mathbf{e}_c$ , the cluster-robust variance is

$$\widehat{\text{Var}}_{\text{cl}} = \mathbf{M} \left( \sum_{c=1}^G \mathbf{u}_c \mathbf{u}_c^T \right) \mathbf{M}, \quad \mathbf{u}_c = \mathbf{Z}_c^T (\mathbf{I} - \mathbf{H}_{cc})^{-1/2} \mathbf{e}_c, \quad (6)$$

with  $\mathbf{H}_{cc} = \mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^T$  (Moore–Penrose inverse root where  $\mathbf{I} - \mathbf{H}_{cc}$  is singular). This is the bias-reduced linearisation (CR2) of [41, 42]; without the factor  $(\mathbf{I} - \mathbf{H}_{cc})^{-1/2}$  it is the usual estimator [43, 44], which is biased downwards when the clusters are few, as here. Since  $\mathbf{Z}_c^T f(\mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^T) = f(\mathbf{Z}_c^T \mathbf{Z}_c \mathbf{M}) \mathbf{Z}_c^T$  for  $f(s) = (1 - s)^{-1/2}$ , every  $\mathbf{u}_c$  requires only the eigen-decomposition of a  $2 \times 2$  matrix. The iid variance is  $s^2 \mathbf{M} (\sum_i \mathbf{z}_i \mathbf{z}_i^T) \mathbf{M}$ . Because the number of clusters per tile is small, Eq. (6) is complemented by a split-half estimate: the sites of a city are divided at random into two disjoint halves, the complete estimator (including its own  $\hat{\mu}, \hat{\tau}^2$ ) is refitted on each half, and  $\text{sd}(\hat{\gamma}^{(1)} - \hat{\gamma}^{(2)})/\sqrt{2}$  estimates the standard error of a half-data label; dividing by a further  $\sqrt{2}$  extrapolates to the full data under the assumption that the variance scales with the inverse number of sites. Five random partitions are pooled.

#### 4.3. Tile descriptors

The descriptor must describe exactly the geometry the ray tracer saw. It is therefore computed from the same footprint parts, heights and scene clipping as the extruded mesh, with every part clipped to the tile it falls in. It contains the 32 built-environment features of [8]: footprint ratio and building count in five height classes ( $< 5$ ,  $5\text{--}10$ ,  $10\text{--}30$ ,  $30\text{--}60$ ,  $\geq 60$  m); mean and standard deviation of footprint area, height and volume; total footprint density; and, for horizontal planes at 5, 10, 30, 60 and 100 m, the number of buildings reaching the plane and the mean and standard deviation of the distance between their centroids. The topographic, land-cover and climatic descriptors of [8] are omitted on purpose: the scenes have a flat ground, no water or vegetation surfaces and no weather, so these features would describe nothing the solver modelled and could only act as city identifiers. Descriptors are stored unnormalised; every experiment scales them by the column-wise maximum absolute value of its training cities, never of the target.

#### 4.4. Transfer operators

We compare constant predictors, label-averaging operators and learned regressors.

- *Source median:* the median source label, the no-information reference.
- *3GPP UMa:* the line-of-sight and non-line-of-sight single-slope formulas of TR 38.901 [6] at the reference distance, which give fixed parameters for every tile, and *COST-231 Hata* [45] (metropolitan correction, median site height of the source cities), used outside its validity range of 1.5 GHz to 2 GHz and 1 km to 20 km as a further standard model.
- *Kernel (Nadaraya–Watson)* [46, 47], the operator of [8]:

$$\hat{\theta}_u = \sum_j \tilde{w}_j(u) \theta_j, \quad \tilde{w}_j(u) \propto \exp\left(-\frac{\|\mathbf{v}_u - \mathbf{v}_j\|^2}{2\sigma^2}\right), \quad (7)$$

computed with the minimum squared distance subtracted in the exponent, which leaves the weights unchanged and avoids numerical underflow.

- *GP-ARD*: a Gaussian process with one length scale per descriptor (automatic relevance determination) [48], i.e. a kernel operator whose metric is learned; the hyper-parameters maximise the marginal likelihood on 600 random source tiles (it needs no validation city) and the posterior mean uses every source tile.
- *k-NN*: the mean label of the  $k$  nearest source tiles.
- *Ridge*, a two-layer *MLP*, gradient-boosted trees (*GBDT*, *LightGBM* [49]) and a *neural encoder* with a constrained  $(\gamma, P_{L_0})$  head trained with a heteroscedastic Gaussian likelihood.
- Two domain-adaptation variants that use the unlabelled descriptors of the target city: *GBDT-IW* weights the source tiles by the density ratio of target to source descriptors estimated with a logistic domain classifier, clipped at its 99th percentile [21], and *CORAL* re-colours the source descriptors with the target covariance before ridge regression [23].

Each operator’s hyper-parameters (bandwidth  $\sigma$ ,  $k$ , ridge, CORAL and MLP penalties, number of trees, early-stopping epoch) are chosen on the validation city of the split by the root-mean-square standardised error over both parameters; the chosen values are listed in Table B.17. Stochastic operators are run with five seeds.

*Behaviour of kernel transfer.* Write  $d_j = \|\mathbf{v}_u - \mathbf{v}_j\|^2$ ,  $\delta_j = d_j - \min_k d_k \in [0, D_u]$ ,  $\bar{\theta} = N^{-1} \sum_j \theta_j$  and  $\sigma_\theta^2 = N^{-1} \sum_j \|\theta_j - \bar{\theta}\|^2$ .

**Theorem 1** (Two regimes of kernel transfer). (a) If  $\epsilon_u = D_u/(2\sigma^2) < 1$ , then

$$\hat{\theta}_u = \bar{\theta} - \frac{1}{2\sigma^2} \text{Cov}_j(\delta_j, \theta_j) + O(\epsilon_u^2), \quad \|\hat{\theta}_u - \bar{\theta}\| \leq \frac{D_u \sigma_\theta}{4\sigma^2} + O(\epsilon_u^2), \quad (8)$$

and  $n_{\text{eff}}(u) = N(1 + O(\epsilon_u^2))$ : the operator returns the source mean up to a first-order correction. (b) Let  $g_u$  be the gap between the smallest and second-smallest squared distance. Then the weight of the nearest source tile satisfies

$$\tilde{w}_{(1)} \geq (1 + (N-1)e^{-g_u/(2\sigma^2)})^{-1}, \quad (9)$$

so  $\hat{\theta}_u$  tends to the label of the nearest source tile when  $g_u \gg 2\sigma^2 \ln N$ .

The minimum distance cancels from Eq. (7); a target far from every source tile is therefore *not* by itself pushed to either regime. Which regime applies is decided by the selected bandwidth relative to the spread  $D_u$  and the gap  $g_u$ , and we measure it (Section 5.4) rather than assume it. Both regimes carry a failure mode under shift: the flat regime returns the source mean and ignores the target descriptor; the nearest-neighbour regime copies one source tile and transmits its label noise undamped (Proposition 2 with  $n_{\text{eff}} = 1$ ).

#### 4.5. Few-shot correction

With  $k$  target tiles surveyed, the zero-shot prediction  $g(\mathbf{v})$  is corrected either by the mean anchor residual (*bias*) or by a ridge model of the residual on the descriptors (*residual*, penalty 10 on normalised descriptors). Anchors are drawn uniformly at random, as the  $k$  tiles closest to a random tile (*contiguous*), which mimics a single drive-test area [24], or by a *design* that uses no measurement: the  $k$  tiles that greedily maximise the facility-location objective  $\sum_i \max_{j \in T} \exp(-\|\mathbf{v}_i - \mathbf{v}_j\|^2/2h^2)$  over the target tiles, with  $h$  the median pairwise descriptor distance; the objective is monotone submodular, so the greedy set is within a factor  $(1 - 1/e)$  of the optimum [28, 26]. Evaluation is always on the unsurveyed tiles, anchors are shared by all operators, and ten random draws are made per  $k \in \{3, 5, 10, 25, 50, 100\}$ .

#### 4.6. Site-aware link transfer

Proposition 1 shows that any receiver-tile model assigns every pixel to its nearest site and describes interference through the exponent alone. As a transfer model free of this restriction we predict the path loss of each (pixel, site) link directly. The features are computed from the scene geometry only, so they are available for a city without any measurement. The building heights are rasterised at 2 m and the straight segment from the site to the pixel is sampled at 96 interior points, where the building height  $H(s)$  is compared with the line-of-sight height  $h_{\text{los}}(s)$ . The link features are the 3-D and horizontal distances, the site height and the elevation angle; the fraction of the segment with  $H > h_{\text{los}}$ , the number of separate obstructions and the length over footprints; the largest excess height  $\max_s(H - h_{\text{los}})$ ; the single knife-edge loss  $J(\nu) = 6.9 + 20 \log_{10}(\sqrt{(\nu - 0.1)^2 + 1} + \nu - 0.1)$  for  $\nu > -0.78$  (zero otherwise) of the most obstructing point, with  $\nu = h \sqrt{2(1/d_1 + 1/d_2)/\lambda}$  [50]; whether the receiver lies inside a footprint and that building’s height; and the built fraction and mean building height in a 100 m square window centred on the receiver and the built fraction in a 200 m window centred on the site. The 32 descriptors of the receiver’s tile are appended.

The ray tracer reports only links below the 150 dB cut-off, so a regressor trained on them never learns that a distant, obstructed site is not received at all; it would predict a finite loss for every link and overstate the reach of every site. We therefore use a hurdle model: a gradient-boosted classifier [49] of whether a (pixel, site) pair lies below the cut-off, trained on 120,000 pairs of each training city (all sites and the pixels of its labelled tiles, including the pairs beyond the cut-off), and a gradient-boosted regressor of the path loss, trained on 120,000 ray-traced links of each training city. A link is predicted beyond the cut-off (zero received power) when the classifier gives a probability below one half, and otherwise its path loss is the regression. Both models use early stopping on the validation city (learning rate 0.1, 127 leaves, at most 1500 trees, five seeds). The path-loss error of the link model is that of the regressor over the ray-traced links; the regressor alone, which keeps every link, is reported as an ablation.

*Survey design for site calibration.* In the few-shot variant the links observed in  $k$  surveyed tiles, which carry their serving site as minimisation-of-drive-tests (MDT) reports do [24, 25], calibrate a per-site offset. Model the residual of a link of site  $s$  as  $o_s + \varepsilon$ , with independent site offsets  $o_s$  of prior mean  $\bar{o}$  and variance  $\tau_o^2$  and link noise of variance  $\sigma_\varepsilon^2$ , and let  $\kappa = \sigma_\varepsilon^2/\tau_o^2$ . With  $n_s$  observed links of site  $s$  and mean residual  $\bar{r}_s$ , the posterior mean of  $o_s$  is  $(n_s\bar{r}_s + \kappa\bar{o})/(n_s + \kappa)$ , which we use with  $\bar{o}$  the mean residual of all observed links and  $\kappa = 50$  links, and its posterior variance is  $\tau_o^2\kappa/(n_s + \kappa)$ .

**Proposition 5** (Survey design). *Let  $c_{ts} \geq 0$  be the number of links of site  $s$  observable in tile  $t$  and  $n_s(T) = \sum_{t \in T} c_{ts}$  for a set  $T$  of surveyed tiles. The total reduction of posterior variance of the site offsets,*

$$F(T) = \tau_o^2 \sum_s \frac{n_s(T)}{n_s(T) + \kappa}, \quad (10)$$

*is monotone and submodular in  $T$ , so the greedy choice of  $k$  tiles attains  $F(T_{\text{greedy}}) \geq (1 - 1/e) \max_{|T|=k} F(T)$ .*

Before any measurement  $c_{ts}$  is unknown; the *designed* survey uses the number of links of site  $s$  in tile  $t$  that the hurdle classifier of the zero-shot link model predicts below the cut-off. It is compared with random surveys for  $k \in \{10, 25, 50\}$  (ten draws).

#### 4.7. Network-level evaluation

For the held-out city, each operator’s tile parameters and the true site positions define a predicted path-loss map for every (pixel, site) pair through Eq. (2); the ray-traced map is the truth, with pairs beyond the 150 dB cut-off treated as zero received power. With  $P_{\text{tx}} = 46$  dBm, 20 MHz bandwidth and a 7 dB noise figure ( $N = -94.0$  dBm) we compute the path-loss error over all ray-traced links, the serving site, coverage at  $L \in \{120, 130, 140\}$  dB, the SINR of Eq. (1) and the spectral efficiency (attenuated Shannon, factor 0.6, zero below  $-10$  dB, capped at 4.4 b/s/Hz [32]). An SINR below  $-20$  dB, including a pixel at which no site is predicted to be received, is set to  $-20$  dB (outage) for the truth and every predictor alike. Mobility decisions depend on where the serving cell changes: a pixel is a cell-edge pixel when its eastern or northern neighbour has a different serving site (one pixel per boundary crossing), the *edge displacement* is the mean distance between the true and the predicted cell edges (the average of the two directed mean nearest-edge distances), and the *edge-length error* is the relative error of the number of cell-edge pixels, a proxy for the handover rate along a route.

*Site planning.* Selecting  $K$  of the  $|\mathcal{B}|$  sites to maximise the number of covered pixels is a monotone submodular maximisation, for which the greedy algorithm attains a  $(1 - 1/e)$  approximation of the optimum of the function it is given [28]; site selection with propagation models is a classical planning task [27]. We run the greedy on the *predicted* coverage at 140 dB and score the selected sites on the *true* coverage. The regret is the true coverage of the greedy plan computed on the true map minus that of the plan computed on the predicted map.

For dimensioning, we also count the sites that the greedy order on the predicted map needs to cover 90% of the pixels, and the true coverage those sites achieve.

*Certified coverage.* A planner must decide which pixels to *promise* as covered. For a transfer model and a held-out city  $c$ , let  $r_p = PL_{\text{serv}}(p) - \widehat{PL}_{\text{serv}}(p)$  be the true minus the predicted serving path loss of pixel  $p$  (the smallest path loss over the sites), and  $q_c$  the  $(1 - \alpha)$  quantile of  $\{r_p\}$  over the pixels of  $c$ .

**Proposition 6** (City-level coverage certificate). *Let the  $n$  cities be exchangeable and the training procedure symmetric in its source cities (the validation city drawn at random). With  $q_c$  computed from the leave-one-city-out model of each city  $c$ , let  $\hat{q}_t$  be the  $j$ -th largest of  $\{q_c\}_{c \neq t}$  for a target city  $t$ , and claim pixel  $p$  as covered at level  $L$  when  $\widehat{PL}_{\text{serv}}(p) + \hat{q}_t \leq L$ . Then, with probability at least  $(n - j)/n$  over the draw of the target city, at most a fraction  $\alpha$  of its pixels are claimed but not covered.*

The guarantee is at the level of cities, the unit of shift, in the spirit of conformal prediction [29, 30]; it needs no model of the residuals. The rank  $j$  trades confidence for coverage:  $j = 1$  (the largest margin of the other cities) gives probability  $(n - 1)/n$ ,  $j = 2$  gives  $(n - 2)/n$  with a smaller margin. We report it at  $L = 140$  dB for  $\alpha \in \{0.05, 0.10\}$  and  $j \in \{1, 2\}$  (the results at 130 dB are released with the code) with the claimed and the falsely claimed share of the pixels, and compare it with the naive claim  $\widehat{PL}_{\text{serv}}(p) \leq L$ .

#### 4.8. Robustness analyses

*Mixed-path composition.* The framework of [8], following MAPLE [7], composes the loss of a link from every tile its straight path crosses. We split the horizontal segment from each site to each pixel at the tile borders (sampled every 10 m), so that tile  $t$  holds a length  $L_t$  of it, between the distances  $d_t^{\text{in}}$  and  $d_t^{\text{out}}$  from the site, and evaluate three compositions. The first is the sum as written in the published framework,  $P_L = \sum_t [A_t + 10\gamma_t \log_{10} L_t]$  with  $L_t$  in metres, in which every crossed tile adds its own log-distance loss. The second reads the same algorithm as a multi-slope law along the path,  $P_L = A_s + \sum_t 10\gamma_t \log_{10} (\max(d_t^{\text{out}}, d_0) / \max(d_t^{\text{in}}, d_0))$ , with the intercept of the site’s tile  $s$  and, over each stretch of distance, the exponent of the tile it lies in (piecewise). The third averages the parameters along the path,  $P_L = \sum_t (L_t/d_h) [A_t + \gamma_t x]$  with the horizontal length  $d_h$ . The piecewise and path-averaged compositions reduce to Eq. (2) when the whole path lies in one tile. All three are linear in the tile parameters, which are fitted jointly to all links of a city by least squares with a weak ridge penalty ( $\lambda = 10$ ) towards the best city-wide pair; they are evaluated with tiles of 100 m and, as in [8], of 200 m, 500 m and 1000 m, always on the pixels of the main corpus. The labels of the tiles crossed by at least 30 links are transferred with the source median, kernel transfer and GBDT under the protocol of Section 4.9 (one seed), and the predicted maps are scored as in Section 4.7. *Network size.* Every zero-shot predictor is also scored on the network of the first ten sites selected by the siting rule (one site per 0.9 km<sup>2</sup>, inter-site distance about

1020 m). *Siting policy.* The models trained on the main corpus are applied unchanged to the siting replicate; the tile model is scored with labels fitted on the replicate itself and with the labels of the same tiles from the main corpus. *Tile size.* Labels, descriptors, zero-shot transfer (median, ridge, GBDT) and the network evaluation are repeated for  $G_s \in \{50, 100, 200\}$  m. *Censoring.* Links beyond the cut-off are missing, which can bias a least-squares slope. We refit the labels with the cut lowered to 140 and 130 dB and, for 300 random tiles per city, compare the least-squares slope with that of a censored-Gaussian (Tobit) model [31] that also uses the (pixel, site) pairs beyond the cut. *Second band.* The tile and link pipelines are repeated at 7.5 GHz, and the 3.5 GHz link models are applied to the second band with the free-space offset  $20 \log_{10}(f_2/f_1)$ , their knife-edge feature evaluated at the new wavelength. *External scenes.* The Munich, Étoile and Florence scenes distributed with Sionna RT were modelled independently of our corpus, with detailed meshes and several ITU materials (brick, concrete, marble, metal, wood). We derive building footprints and heights from their meshes by vertical ray casting on a 2 m grid, place 8 rooftop sites per scene with the rule of our corpus (minimum separation 150 m, the scenes being smaller), ray trace them with the corpus settings and evaluate, without any retraining, the tile labels, the source median and GBDT fitted on all eleven cities, and the mean of the eleven leave-one-city-out link models, zero-shot and with the per-site calibration of Section 4.6 from  $k \in \{10, 25\}$  surveyed tiles (ten random draws, and the design of Proposition 5). The San Francisco scene has terrain and would require a terrain-following measurement surface; it is not used.

#### 4.9. Protocol and inference

Every city is the target once (leave-one-city-out); one of the remaining cities, drawn with a fixed seed, is the validation city and the others are training cities. Tiles of one city are spatially correlated, so tile-level tests would overstate significance; the unit of inference is the target city. For each comparison we report the mean per-city difference, a 95% percentile bootstrap interval over cities [51], the exact two-sided Wilcoxon signed-rank  $p$ -value [52] and the number of cities in which the first operator is better, with Holm’s correction [53] within each family of comparisons. Standard deviations across cities use  $n - 1$ .

## 5. Results

### 5.1. Corpus and ray-budget convergence

The corpus contains 11 cities, 105,800 buildings, 220 sites and 8,977,400 ray-traced links (Table 1); 10,428 tiles have at least 30 links and are labelled. Figure 2 shows one city. The share of buildings whose height is imputed from the city median ranges from 2% (New York) to 50% (London).

The radio-map solver is a Monte-Carlo method: a cell that no ray reaches reports zero power, so the set of reached cells and their values depend on the ray budget. Table 2 compares budgets with a reference run of  $3 \times 10^8$  rays for one site. At the  $1 \times 10^8$  rays per site used for the corpus, path loss below

120 dB agrees with the reference to 0.53 dB RMS and path loss below 140 dB to 2.0 dB with a mean difference of 0.05 dB, and the number of cells reached below 140 dB is 96.1% of the reference. The complete corpus took 4.7 h on a CPU (LLVM) backend (77 s per site).

### 5.2. Tile labels and their precision

Table 3 summarises the labels. City medians of the tile exponent range from 2.39 (Vienna) to 3.15 (Hamburg); their standard deviation across cities, 0.27, is only 0.3 times the typical standard deviation of the tiles within a city (0.60–0.96). Most of the variation of the tile exponent is therefore within cities, not between them. The fit residual of the log-distance law inside a tile is 11.2–12.7 dB.

Without shrinkage, the OLS exponents span  $-4.8$  to  $8.8$ , the ill-conditioning predicted by Proposition 3; empirical-Bayes shrinkage (median weight 0.36) narrows this to  $-0.90$  to  $6.45$ . Label precision is the decisive quantity (Table 4). The standard error assuming independent residuals is 0.141; the cluster-robust error with sites as clusters is 0.456, 3.2 times larger, and the split-half estimate extrapolated to the full data is 0.686. The exponent of a tile therefore depends strongly on which sites serve it, and a label error of this size is comparable to the differences between cities.

The descriptors carry signal but no single one explains much of the variation (Fig. 3b): the strongest correlation with the exponent is  $r = -0.45$ , for footprint density.

### 5.3. Zero-shot transfer of tile parameters

Table 5 and Fig. 4 report the exponent error on every held-out city. The source median attains 0.871; kernel transfer 0.813; the Gaussian process with learned length scales 0.764; the learned regressors 0.769 (ridge), 0.761 (MLP), 0.767 (neural encoder) and 0.782 (GBDT). The domain-adaptation variants do not improve on their base learners (CORAL 0.787 against ridge 0.769; GBDT-IW 0.784 against GBDT 0.782), and the standard models are worse than the median (3GPP UMa 1.003 for line of sight and 1.453 for non-line of sight, COST-231 Hata 1.054). The lowest mean error is that of the MLP, but the learned operators differ by less than their spread across cities. For the intercept at 100 m the learned operators reduce the error from 13.8 dB (median) to 9.6 dB (GBDT).

At the level of cities (Table D.22), GBDT improves on the median by  $-0.089$  (95% CI  $[-0.132, -0.043]$ ,  $p = 0.010$ , better in 9 of 11 cities) and on kernel transfer by  $-0.030$  ( $[-0.057, -0.008]$ ); kernel transfer does not improve significantly on the median ( $-0.059$ ,  $[-0.111, +0.005]$ ,  $p = 0.147$ ), whereas learning the metric of the same kernel does: GP-ARD improves on kernel transfer by  $-0.049$  ( $[-0.075, -0.025]$ ,  $p = 0.005$ , better in 9 of 11 cities), which remains significant after Holm’s correction. With eleven cities, however, none of the improvements over the source median is significant after Holm’s correction over its 12 comparisons; the only significant differences are deteriorations (3GPP UMa NLoS). The hardest target is Chicago, where no transfer operator improves on the source median (0.835).

Table 1: Corpus. Height source: share of buildings with an explicit height attribute, with a floor count, or imputed from the city median. Links: ray-traced (pixel, site) pairs below the 150 dB cut-off. Tiles: labelled tiles ( $\geq 30$  links).

| City      | Buildings | Height source (%) |         |         | Median (m) | Links     | Tiles  |
|-----------|-----------|-------------------|---------|---------|------------|-----------|--------|
|           |           | tag               | storeys | imputed |            |           |        |
| Amsterdam | 21,135    | 83                | 2       | 15      | 15.5       | 736,255   | 948    |
| Barcelona | 12,337    | 12                | 64      | 23      | 19.2       | 609,050   | 942    |
| Berlin    | 6,036     | 15                | 63      | 22      | 16.0       | 1,009,619 | 957    |
| Chicago   | 2,952     | 70                | 4       | 25      | 15.9       | 1,069,552 | 961    |
| Hamburg   | 4,335     | 17                | 42      | 41      | 16.0       | 1,152,125 | 961    |
| London    | 11,891    | 8                 | 42      | 50      | 12.8       | 931,288   | 940    |
| Munich    | 8,616     | 9                 | 49      | 42      | 12.8       | 794,056   | 950    |
| New York  | 7,126     | 98                | 1       | 2       | 18.6       | 827,510   | 960    |
| Paris     | 14,383    | 3                 | 64      | 33      | 19.2       | 529,649   | 910    |
| Prague    | 9,907     | 1                 | 70      | 29      | 16.0       | 783,805   | 960    |
| Vienna    | 7,082     | 26                | 60      | 14      | 16.0       | 534,491   | 939    |
| Total     | 105,800   |                   |         |         |            | 8,977,400 | 10,428 |

Table 2: Ray-budget convergence for one site (Amsterdam, 10 m cells). For each path-loss level: number of cells reached below that level, relative to the  $3 \times 10^8$ -ray reference (%; Monte-Carlo noise can exceed 100%), and RMS path-loss difference to the reference over the cells reached by both (dB). s: solver time.

| Rays            | s   | $\leq 120$ dB |      | $\leq 130$ dB |      | $\leq 140$ dB |      | $\leq 150$ dB |      |
|-----------------|-----|---------------|------|---------------|------|---------------|------|---------------|------|
|                 |     | cells         | RMS  | cells         | RMS  | cells         | RMS  | cells         | RMS  |
| $1 \times 10^7$ | 9   | 95.9          | 1.69 | 82.4          | 2.92 | 75.1          | 4.48 | 74.7          | 6.68 |
| $3 \times 10^7$ | 27  | 99.8          | 0.98 | 92.7          | 1.88 | 87.7          | 3.11 | 86.1          | 4.54 |
| $1 \times 10^8$ | 98  | 100.3         | 0.53 | 98.4          | 1.14 | 96.1          | 2.00 | 94.9          | 3.13 |
| $3 \times 10^8$ | 245 | 100.0         | 0.00 | 100.0         | 0.00 | 100.0         | 0.00 | 100.0         | 0.00 |

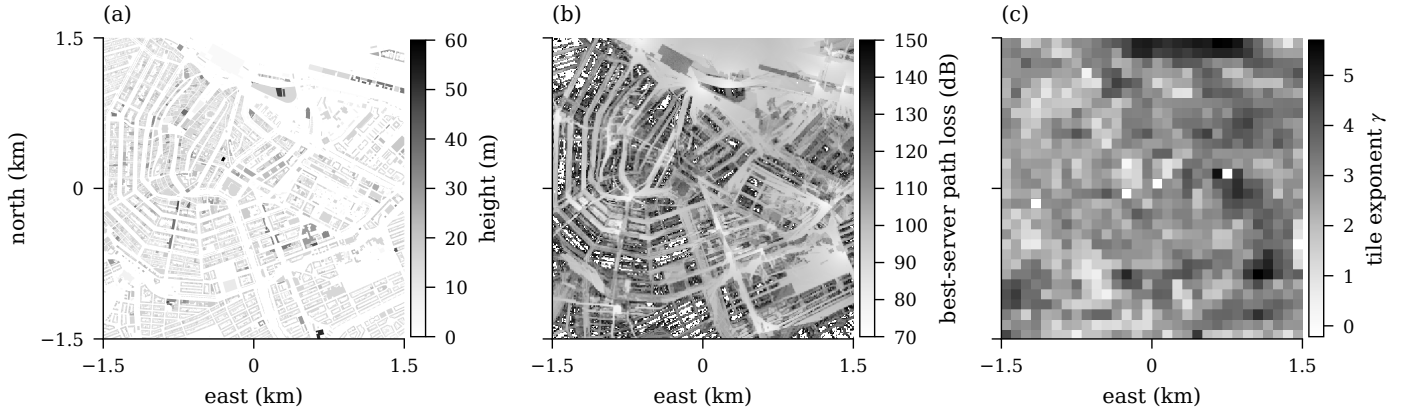


Figure 2: Amsterdam. (a) Building heights of the scene. (b) Ray-traced path loss to the best site. (c) Tile exponents  $\hat{\gamma}_t$ .

#### 5.4. Regime of kernel transfer

Table 6 places the tuned kernel operator relative to Theorem 1. The selected bandwidths are  $\sigma = 0.1$ – $0.5$  and the median  $\epsilon_u$  ranges from 11.4 to 315: no target tile satisfies the condition of part (a), and the largest single weight has median 0.01, so the operator is not a nearest-neighbour copy either. Where the largest bandwidth is selected, however, it averages over up to 73% of the source tiles and its predictions spread over as little as 0.18 of the spread of the target labels: it approaches the source mean although part (a) does not hold. Across the targets the prediction spread is 0.18–0.75 of that of the labels. Its disadvantage is the noise-transmission term of Proposition 2

combined with an isotropic distance in which every descriptor counts equally; the Gaussian process, which learns one length scale per descriptor, removes most of the gap to the learned regressors (Section 5.3).

#### 5.5. Few-shot correction of tile parameters

Surveying target tiles does not help the tile exponent at small  $k$  (Table 7, Fig. 5). With ten random tiles the residual correction changes the GBDT error from 0.782 to 0.786 and improves it in only 3 of 11 cities; for ridge the change is  $+0.013$  ( $[+0.004, +0.020]$ ). The anchors' labels carry the site-dependent error of Section 5.2, so a correction fitted to ten of them adds more

Table 3: Tile labels per city. City-wide fit: OLS over all links. Tile labels: median and standard deviation of  $\hat{\gamma}_t$  and median  $\hat{P}_{L_0,t}$  (dB, at  $d_0 = 100$  m);  $\hat{\tau}$ : between-tile standard deviation of the slope;  $w_t$ : shrinkage weight; RMSE: link-weighted fit residual (dB).

| City      | City-wide fit |           | Tile labels   |               |                | $\hat{\tau}$ | med. $w_t$ | RMSE |
|-----------|---------------|-----------|---------------|---------------|----------------|--------------|------------|------|
|           | $\gamma$      | $P_{L_0}$ | med. $\gamma$ | s.d. $\gamma$ | med. $P_{L_0}$ |              |            |      |
| Amsterdam | 2.47          | 100.5     | 2.84          | 0.93          | 100.9          | 1.06         | 0.27       | 11.3 |
| Barcelona | 1.92          | 105.4     | 2.43          | 0.96          | 105.5          | 1.14         | 0.28       | 11.6 |
| Berlin    | 2.65          | 98.2      | 2.82          | 0.81          | 97.5           | 1.00         | 0.36       | 11.6 |
| Chicago   | 0.81          | 115.6     | 2.79          | 0.83          | 97.2           | 1.19         | 0.54       | 12.7 |
| Hamburg   | 2.14          | 100.2     | 3.15          | 0.72          | 92.2           | 0.94         | 0.37       | 11.4 |
| London    | 1.72          | 104.4     | 2.40          | 0.60          | 103.3          | 0.79         | 0.42       | 11.2 |
| Munich    | 2.12          | 104.0     | 3.00          | 0.80          | 99.8           | 0.92         | 0.33       | 11.2 |
| New York  | 1.11          | 115.9     | 2.69          | 0.79          | 102.2          | 1.18         | 0.53       | 12.6 |
| Paris     | 2.06          | 103.6     | 2.41          | 0.95          | 106.7          | 1.13         | 0.29       | 11.8 |
| Prague    | 2.48          | 100.9     | 2.92          | 0.77          | 99.0           | 0.95         | 0.37       | 11.4 |
| Vienna    | 2.29          | 106.1     | 2.39          | 0.92          | 108.4          | 1.09         | 0.32       | 11.4 |

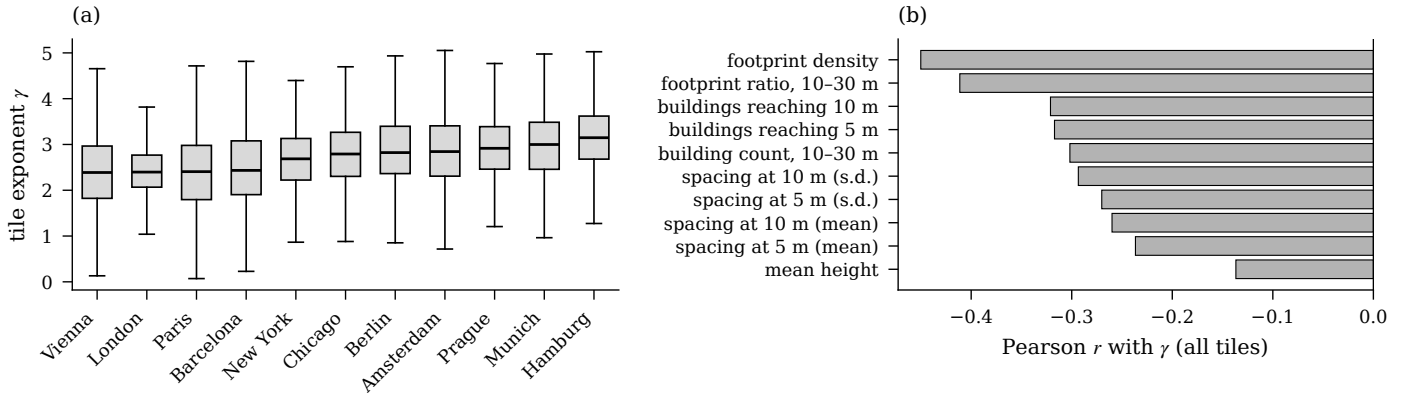


Figure 3: (a) Tile exponents per city (box: quartiles; whiskers: 1.5 times the interquartile range). (b) The ten descriptors most correlated with the tile exponent over all 10,428 tiles.

Table 4: Standard error of the tile labels: independent residuals (iid), cluster-robust over sites (cluster), and split-half over random halves of the sites extrapolated to all sites (split). Root mean square over tiles.

| City      | sites/tile | se( $\gamma$ ) |         |       | se( $P_{L_0}$ ) (dB) |         |       |
|-----------|------------|----------------|---------|-------|----------------------|---------|-------|
|           |            | iid            | cluster | split | iid                  | cluster | split |
| Amsterdam | 18         | 0.140          | 0.440   | 0.600 | 1.42                 | 5.06    | 6.90  |
| Barcelona | 18         | 0.178          | 0.470   | 0.592 | 1.73                 | 5.31    | 6.91  |
| Berlin    | 19         | 0.131          | 0.445   | 0.583 | 1.50                 | 5.34    | 6.99  |
| Chicago   | 19         | 0.117          | 0.545   | 1.006 | 1.33                 | 6.73    | 12.38 |
| Hamburg   | 20         | 0.098          | 0.402   | 0.629 | 1.09                 | 4.77    | 7.45  |
| London    | 19         | 0.105          | 0.353   | 0.537 | 1.18                 | 4.39    | 6.62  |
| Munich    | 18         | 0.131          | 0.397   | 0.559 | 1.41                 | 4.76    | 6.75  |
| New York  | 17         | 0.142          | 0.541   | 0.890 | 1.54                 | 6.65    | 10.40 |
| Paris     | 17         | 0.187          | 0.488   | 0.651 | 1.91                 | 5.57    | 7.56  |
| Prague    | 19         | 0.115          | 0.416   | 0.667 | 1.25                 | 4.84    | 7.45  |
| Vienna    | 16         | 0.179          | 0.481   | 0.674 | 1.80                 | 5.42    | 7.69  |
| RMS       |            | 0.141          | 0.456   | 0.686 | 1.49                 | 5.40    | 8.10  |

variance than it removes. The error falls below the zero-shot value only from  $k = 25$  surveyed tiles on (GBDT 0.760), and reaches 0.743 at  $k = 100$ . Contiguous survey areas are worse than random tiles at every  $k$  (GBDT at  $k = 10$ : 0.882), because neighbouring tiles repeat the same information. The facility-location design, which spreads the survey over the descriptor space without any measurement, gives 0.778 at  $k = 10$  ( $-0.007$  against random tiles,  $[-0.022, +0.010]$ , better in 7 of 11 cities);

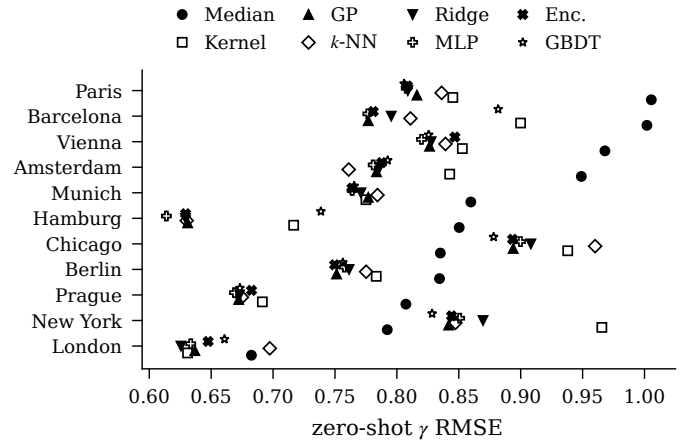


Figure 4: Zero-shot exponent RMSE per held-out city, ordered by the error of the source median.

from  $k = 10$  on it beats the zero-shot error. With three or five tiles the design is worse than random anchors (Table 7), and at large  $k$  the two converge.

Table 5: Zero-shot exponent RMSE per held-out city (mean of five seeds for stochastic operators; best per row in bold). Last row: mean  $P_{L_0}$  RMSE (dB).

| Target city    | Median       | UMa-L | UMa-N | COST  | Kernel | GP    | $k$ -NN      | Ridge | CORAL        | MLP          | Enc.         | GBDT         | IW    |
|----------------|--------------|-------|-------|-------|--------|-------|--------------|-------|--------------|--------------|--------------|--------------|-------|
| Amsterdam      | 0.949        | 1.146 | 1.394 | 1.035 | 0.843  | 0.784 | <b>0.761</b> | 0.786 | 0.820        | 0.781        | 0.788        | 0.793        | 0.801 |
| Barcelona      | 1.002        | 0.998 | 1.730 | 1.318 | 0.900  | 0.777 | 0.811        | 0.795 | 0.817        | <b>0.777</b> | 0.781        | 0.882        | 0.895 |
| Berlin         | 0.834        | 1.052 | 1.319 | 0.946 | 0.783  | 0.751 | 0.775        | 0.761 | 0.877        | 0.758        | <b>0.750</b> | 0.756        | 0.756 |
| Chicago        | <b>0.835</b> | 1.006 | 1.414 | 1.031 | 0.938  | 0.894 | 0.960        | 0.908 | 0.876        | 0.900        | 0.893        | 0.878        | 0.871 |
| Hamburg        | 0.850        | 1.162 | 1.078 | 0.770 | 0.717  | 0.631 | 0.630        | 0.629 | 0.616        | <b>0.614</b> | 0.629        | 0.739        | 0.668 |
| London         | 0.683        | 0.639 | 1.599 | 1.097 | 0.631  | 0.637 | 0.697        | 0.626 | <b>0.624</b> | 0.634        | 0.648        | 0.661        | 0.669 |
| Munich         | 0.860        | 1.120 | 1.229 | 0.875 | 0.775  | 0.777 | 0.784        | 0.771 | 0.806        | 0.764        | <b>0.764</b> | 0.766        | 0.766 |
| New York       | <b>0.792</b> | 0.907 | 1.489 | 1.064 | 0.966  | 0.842 | 0.847        | 0.870 | 0.860        | 0.851        | 0.844        | 0.828        | 0.877 |
| Paris          | 1.005        | 0.977 | 1.748 | 1.308 | 0.845  | 0.816 | 0.836        | 0.809 | 0.818        | 0.808        | 0.809        | <b>0.806</b> | 0.811 |
| Prague         | 0.807        | 1.086 | 1.219 | 0.851 | 0.691  | 0.672 | 0.675        | 0.673 | 0.733        | <b>0.669</b> | 0.683        | 0.673        | 0.683 |
| Vienna         | 0.968        | 0.938 | 1.761 | 1.302 | 0.853  | 0.826 | 0.839        | 0.828 | <b>0.805</b> | 0.820        | 0.847        | 0.826        | 0.824 |
| Mean           | 0.871        | 1.003 | 1.453 | 1.054 | 0.813  | 0.764 | 0.783        | 0.769 | 0.787        | <b>0.761</b> | 0.767        | 0.782        | 0.784 |
| S.d.           | 0.100        | 0.147 | 0.234 | 0.191 | 0.105  | 0.085 | 0.092        | 0.092 | 0.091        | 0.089        | 0.084        | 0.073        | 0.083 |
| $P_{L_0}$ (dB) | 13.8         | 21.6  | 13.8  | 17.9  | 10.5   | 9.5   | 9.8          | 9.7   | 10.3         | 9.5          | 9.6          | 9.6          | 9.6   |

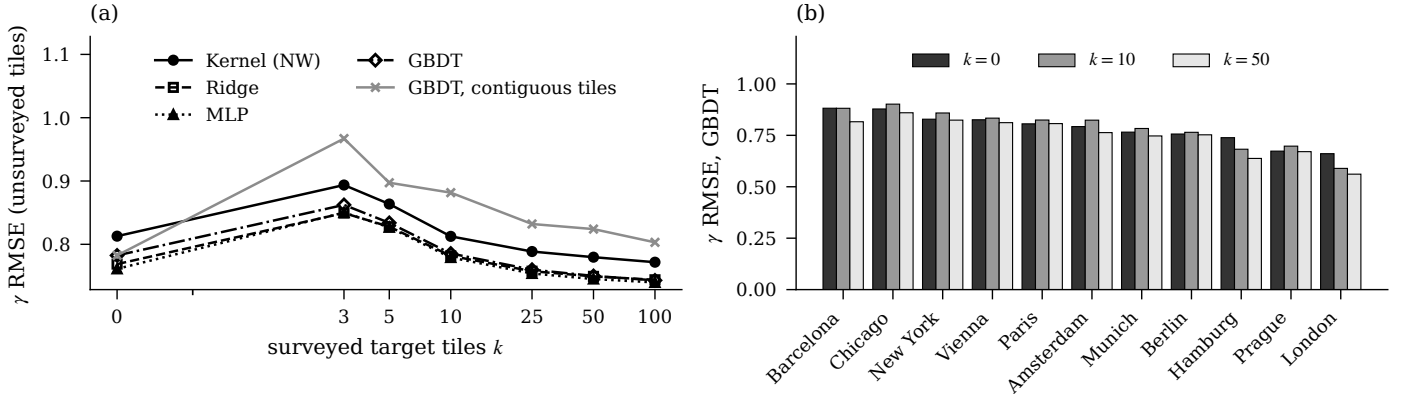


Figure 5: (a) Exponent RMSE against the number of surveyed target tiles. (b) GBDT per held-out city at  $k \in \{0, 10, 50\}$ .

Table 6: Kernel transfer on each held-out city: selected bandwidth; median  $\epsilon_u = D_u/2\sigma^2$  and share of tiles with  $\epsilon_u < 1$  (Theorem 1a); median effective-neighbourhood fraction; median largest weight (Theorem 1b); standard deviation of the predictions over that of the labels.

| Target    | $\sigma$ | med. $\epsilon_u$ | $\epsilon_u < 1$ (%) | $n_{\text{eff}}/N$ (%) | med. $w_{\text{max}}$ | spread |
|-----------|----------|-------------------|----------------------|------------------------|-----------------------|--------|
| Amsterdam | 0.1      | 315               | 0                    | 2.19                   | 0.020                 | 0.49   |
| Barcelona | 0.5      | 11.4              | 0                    | 70.90                  | 0.000                 | 0.18   |
| Berlin    | 0.1      | 294               | 0                    | 5.07                   | 0.009                 | 0.44   |
| Chicago   | 0.1      | 307               | 0                    | 0.45                   | 0.073                 | 0.65   |
| Hamburg   | 0.5      | 12.4              | 0                    | 73.05                  | 0.000                 | 0.27   |
| London    | 0.2      | 75.8              | 0                    | 22.15                  | 0.001                 | 0.60   |
| Munich    | 0.2      | 75.3              | 0                    | 27.00                  | 0.001                 | 0.38   |
| New York  | 0.1      | 277               | 0                    | 0.04                   | 0.467                 | 0.75   |
| Paris     | 0.1      | 304               | 0                    | 5.00                   | 0.008                 | 0.39   |
| Prague    | 0.1      | 299               | 0                    | 6.28                   | 0.007                 | 0.43   |
| Vienna    | 0.1      | 294               | 0                    | 5.54                   | 0.008                 | 0.43   |

## 5.6. Network-level evaluation

Table 8 and Fig. 6 evaluate the predictors where a planner uses them. The first row is the tile model with the true labels (oracle), which isolates the error of the tile model itself. The oracle has a path-loss error of 11.6 dB, selects the correct serving site for 50.1% of the pixels (the nearest-site rule of Proposi-

Table 7: Few-shot correction: exponent RMSE on the unsurveyed tiles, mean over cities (and ten anchor draws for random and contiguous anchors). Left: residual correction with random anchors. Right: GBDT with a bias correction, with contiguous anchors and with the facility-location design.

| $k$ | residual correction, random tiles |       |       |       | GBDT  |         |        |
|-----|-----------------------------------|-------|-------|-------|-------|---------|--------|
|     | Kernel                            | Ridge | MLP   | GBDT  | bias  | contig. | design |
| 0   | 0.813                             | 0.769 | 0.761 | 0.782 | 0.782 | 0.782   | 0.782  |
| 3   | 0.894                             | 0.849 | 0.850 | 0.862 | 0.864 | 0.967   | 0.905  |
| 5   | 0.864                             | 0.828 | 0.827 | 0.834 | 0.836 | 0.897   | 0.840  |
| 10  | 0.813                             | 0.782 | 0.779 | 0.786 | 0.790 | 0.882   | 0.778  |
| 25  | 0.789                             | 0.758 | 0.754 | 0.760 | 0.770 | 0.832   | 0.759  |
| 50  | 0.780                             | 0.749 | 0.745 | 0.750 | 0.765 | 0.824   | 0.755  |
| 100 | 0.772                             | 0.744 | 0.740 | 0.743 | 0.762 | 0.803   | 0.749  |

tion 1) and misjudges the SINR by 7.6 dB on average. Transferred tile parameters reduce the path-loss error of the source median from 14.5 dB to 12.8 dB (GBDT;  $-1.72$  dB, better in 11 of 11 cities) but remain above the oracle, and a plan of  $K = 5$  sites made on their map loses 20.6% of the pixels against 9.3% for the oracle.

Site-aware link transfer removes the ceiling of the tile model. Its path-loss error is 9.3 dB (8.8 to 9.6 dB across cities), 3.5 dB

below tile transfer with GBDT (95% CI  $[-3.90, -3.12]$ , better in 11 of 11 cities) and 2.3 dB below the tile model with the *true* labels ( $[-2.61, -2.07]$ , 11 of 11). It selects the correct serving site for 68.8% of the pixels (+18.7 percentage points against GBDT, in 11 of 11 cities), misjudges the SINR by 5.4 dB on average and loses 0.5% of the pixels when planning  $K = 5$  sites, against 20.6% for GBDT tile transfer and 9.3% for the true tile labels. The hurdle is what makes the plans reliable: the regressor alone, which keeps every link, has the same path-loss error but a planning regret of 5.6%, because it lets distant, obstructed sites cover pixels they do not reach.

*Cell edges and dimensioning.* Table 9 turns the maps into mobility and dimensioning quantities. Every tile model draws the nearest-site cell pattern of Proposition 1: its cell edges lie 80 m from the true edges on average (76 m with the true labels), and their total length is in error by  $-93\%$ : the ray-traced serving pattern is fragmented by buildings and a route crosses many more cell edges than the nearest-site pattern suggests. Link transfer places the edges 7 m from the truth with an edge-length error of  $-5\%$ . For dimensioning, the truth needs on average 6.7 of the 20 sites to cover 90% of the pixels at 140 dB. The tile models, which cannot represent the obstruction of distant sites, predict that a single site suffices (GBDT; 1.1 sites with the true labels, averages over the cities), and the sites they choose cover only 47.4% of the pixels in truth. Link transfer chooses 3.8 sites on average, which cover 81.6%; it still under-dimensions the network, but by far less (the regressor alone: a single site, 56.9%).

The model relies on the propagation features, which account for 97% of the gain of its trees (Table 10); the single most important feature is the knife-edge loss (23%), and the three most important together account for 60%.

### 5.7. Mixed-path composition

Composing the tile parameters along each path removes the nearest-site restriction of Proposition 1, and with exact labels it helps (Table 11): the path-averaged composition, fitted to the links of each city, has a path-loss error of 11.1 dB against 11.6 dB for the receiver-tile labels, selects the correct serving site for 54.1% of the pixels against 50.1%, places the cell edges 25 m from the true ones and loses 5.3% of the pixels when planning five sites. Most of the gain does not survive transfer. With parameters transferred by GBDT the path-averaged composition has a path-loss error of 13.4 dB, above that of receiver-tile transfer (12.8 dB), and a serving-site accuracy of 46.3%, below the nearest-site rule in all but 2 of 11 cities: every link now accumulates the transfer errors of all the tiles it crosses, and these errors decide which site appears strongest. Only its planning regret improves, to 18.3% against 20.6%, still far above the 0.5% of link transfer. The piecewise composition gains nothing over the receiver tile even with exact labels (11.6 dB, 49.1%), and with transferred exponents it is the worst of the three (GBDT 16.9 dB, 34.3%): an error in the exponent of one tile shifts the loss of every link that passes through it. The summed composition of [8] does not describe the ray-traced links even with parameters fitted to the city itself (30.0 dB to 36.2 dB): every

crossed tile adds its own loss  $A_t + 10\gamma_t \log_{10} L_t$ , so the loss grows with the number of tiles crossed rather than with the logarithm of the distance, and the transferred parameters give errors of 74.5 to 92.7 dB. Larger tiles, as in [8], do not change the picture (Table C.21, Appendix Appendix C): with tiles of 200 m to 1000 m the summed composition with exact labels has errors of 40.0 to 42.7 dB, larger than with 100 m tiles, because fewer tile parameters must absorb the same mismatch, and the path-averaged and piecewise compositions lose accuracy as the tiles grow (11.8 to 13.0 dB and 12.2 to 13.1 dB with exact labels). Over all compositions and tile sizes, the best tile model with exact labels reaches 11.1 dB and 54.1% of the serving sites; site-aware link transfer is better than every composition with exact labels in path loss and serving site, and at least as good in planning regret, in every held-out city. The limitation lies in describing a link by tile parameters, however they are combined and whatever the tile size.

### 5.8. Survey design for site calibration

Calibrating the per-site offsets of the link model from surveyed tiles barely changes it on the corpus (Table 12). With ten random tiles the path-loss error rises from 9.33 dB to 9.46 dB (+0.13 dB, worse in every city); only fifty random tiles improve it, marginally, to 9.30 dB with 69.1% of the serving sites correct, and with fifty designed tiles it is 9.44 dB; the design of Proposition 5, which maximises the variance reduction of the offsets, is not better than random tiles. This indicates that the zero-shot link model has little systematic per-site error left for a survey of this size to remove, while an offset estimated from the links of a few tiles transmits their local residuals to the whole cell. Calibration pays off where a systematic error exists: on the external scenes of Section 5.10, whose materials shift the path loss, it removes the bias. The design is worse than random surveys there as well, and its offsets are biased (Table 16): maximising the variance reduction favours tiles that see many sites, which need not be representative of the city, so an unbiased (random or stratified) survey is preferable.

### 5.9. Certified coverage

Table 13 applies Proposition 6. For every predictor the bound  $\hat{q}_t$  holds in 10 of 11 cities. This count is fixed by construction (with distinct margins, the bound of rank  $j$  fails exactly in the  $j$  cities with the largest  $q_c$ ) and is a consistency check, not evidence; the guarantee concerns a new city drawn exchangeably with the eleven. At  $\alpha = 0.1$  and 140 dB, the certificate of link transfer needs a margin of only 9.8 dB and certifies 92.0% of the pixels, with 1.79% falsely claimed; the true covered share is 96.3%. Tile transfer needs larger margins and certifies less: GBDT 78.3% (margin 17.6 dB), the source median 60.5% and even the true tile labels 88.2%. The naive claim without a margin covers almost every pixel but is wrong for 3.11% (link) to 3.68% (GBDT) of them, with no stated risk. Using the second-largest margin of the other cities ( $j = 2$ ) lowers the confidence to 9/11 and raises the certified share of link transfer to 93.5% (GBDT 83.2%); the bound then holds in 9 of 11 cities. The falsely claimed shares stay far below  $\alpha$  because the margin must

Table 8: Network-level evaluation on the held-out cities (mean over cities). Oracle: tile model with the true labels. Serving site: share of pixels with the correct serving site. Coverage: agreement of covered pixels (serving path loss  $\leq 130$  dB) and error of the covered share. SINR: mean absolute error and Kolmogorov–Smirnov distance of the distributions. SE: error of mean spectral efficiency (b/s/Hz). Regret: coverage lost by planning  $K = 5$  sites on the predicted map instead of the true map. True covered share: 89.3%; true mean spectral efficiency: 1.47 b/s/Hz.

| Predictor                       | Path loss (dB) |        | Serving site (%) | Coverage 130 dB (%) |          | SINR     |       | SE       | Regret    |
|---------------------------------|----------------|--------|------------------|---------------------|----------|----------|-------|----------|-----------|
|                                 | RMSE           | bias   |                  | agree               | $\Delta$ | MAE (dB) | KS    | $\Delta$ | $K=5$ (%) |
| Tile labels (oracle)            | 11.64          | +0.00  | 50.1             | 89.6                | 6.0      | 7.56     | 0.304 | -0.652   | 9.3       |
| Source median                   | 14.52          | +3.68  | 50.1             | 87.9                | 8.9      | 7.18     | 0.290 | -0.668   | 32.1      |
| 3GPP UMa NLoS                   | 22.75          | +17.29 | 50.1             | 70.2                | -16.9    | 6.92     | 0.118 | -0.213   | 8.0       |
| COST-231 Hata                   | 25.29          | +20.89 | 50.1             | 52.8                | -40.1    | 7.14     | 0.218 | -0.494   | 9.9       |
| Kernel (NW)                     | 13.02          | +1.30  | 50.1             | 88.2                | 8.4      | 7.47     | 0.296 | -0.675   | 19.0      |
| GP-ARD                          | 12.80          | +1.02  | 50.1             | 88.1                | 7.5      | 7.56     | 0.298 | -0.676   | 17.9      |
| $k$ -NN                         | 12.83          | +1.12  | 50.1             | 88.4                | 8.5      | 7.56     | 0.291 | -0.665   | 28.8      |
| Ridge                           | 12.80          | +1.06  | 50.1             | 87.9                | 7.6      | 7.54     | 0.298 | -0.674   | 18.9      |
| CORAL + ridge                   | 13.08          | +2.38  | 50.1             | 87.3                | 5.5      | 7.63     | 0.316 | -0.708   | 18.3      |
| MLP                             | 12.76          | +1.02  | 50.1             | 88.2                | 7.8      | 7.57     | 0.297 | -0.673   | 26.5      |
| Neural encoder                  | 12.79          | +0.86  | 50.1             | 88.4                | 8.5      | 7.54     | 0.294 | -0.668   | 25.8      |
| GBDT                            | 12.80          | +0.97  | 50.1             | 88.3                | 8.2      | 7.44     | 0.294 | -0.671   | 20.6      |
| GBDT-IW                         | 12.82          | +0.89  | 50.1             | 88.3                | 8.1      | 7.44     | 0.293 | -0.669   | 18.7      |
| GBDT + 10 tiles                 | 12.75          | +0.96  | 50.1             | 88.5                | 7.7      | 7.56     | 0.299 | -0.673   | 20.4      |
| GBDT + 10 designed tiles        | 12.73          | +1.25  | 50.1             | 88.3                | 7.3      | 7.52     | 0.290 | -0.643   | 19.8      |
| Link transfer (ours)            | 9.33           | -0.01  | 68.8             | 90.4                | 2.4      | 5.37     | 0.048 | -0.078   | 0.5       |
| Link, regressor only            | 9.33           | -0.01  | 68.9             | 90.4                | 2.5      | 5.83     | 0.162 | -0.322   | 5.6       |
| Link + 10 tiles (ours)          | 9.46           | +0.11  | 65.9             | 90.3                | 2.5      | 5.52     | 0.047 | -0.067   | 0.5       |
| Link + 10 designed tiles (ours) | 9.52           | -0.15  | 65.6             | 90.1                | 2.6      | 5.65     | 0.051 | -0.071   | 0.9       |

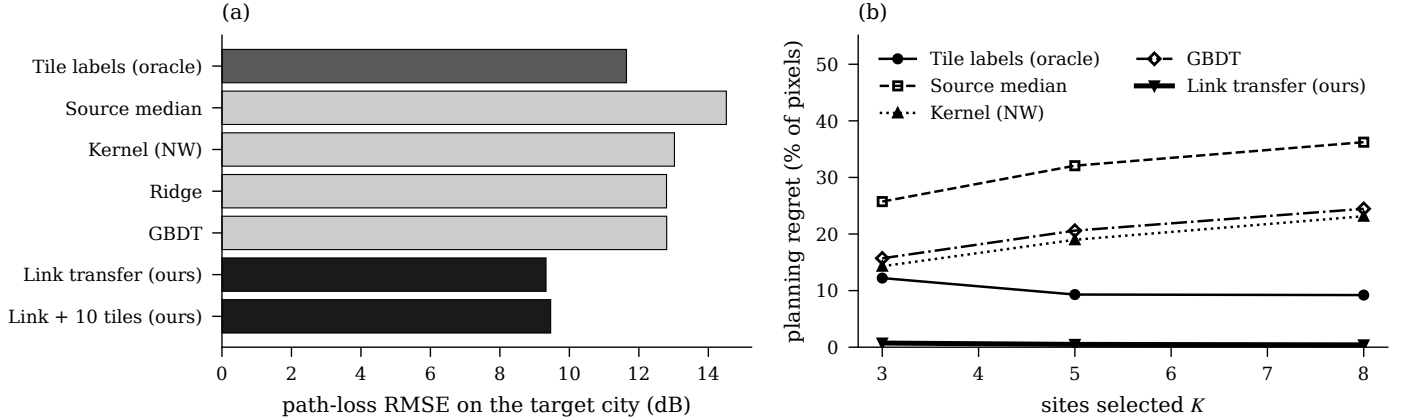


Figure 6: (a) Path-loss RMSE on the held-out cities. (b) Planning regret against the number of selected sites.

cover the worst city; the value of the certificate is the guarantee itself. The accuracy of a transfer model thus translates directly into the coverage an operator can promise at a stated risk.

### 5.10. Robustness

**Tile size.** Table C.18 (Appendix C) repeats the tile pipeline for  $G_s \in \{50, 100, 200\}$  m;  $G_s = 100$  m reproduces the main results exactly. Larger tiles give more precise labels (cluster-robust standard error 0.540, 0.456 and 0.369) and a smaller exponent error (GBDT 0.888, 0.782 and 0.619), but a coarser map: the path-loss error of the oracle grows from 11.2 dB to 12.1 dB, and that of GBDT tile transfer stays between 12.7 dB and 13.1 dB. The serving-site accuracy of the tile model is 50.1% at every

size, as Proposition 1 implies. No tile size approaches link transfer.

**Path-loss cut-off.** Between 35.5% and 66.4% of the (pixel, site) pairs of the labelled tiles lie beyond the cut-off (Table C.19, Appendix C). The labels depend on this dynamic range: lowering the cut-off to 140 dB changes the median tile exponent by  $-0.583$  and to 130 dB by  $-1.119$ , and a Tobit fit that also uses the censored pairs gives slopes larger than the least-squares slopes by a median of 2.11 to 4.49 across the cities (3,300 tiles). The tile exponent thus depends on the dynamic range of the measurements, not only on the environment, which is a further reason to model links, and to model the cut-off explicitly as the hurdle of the link model does.

Table 9: Cell edges and dimensioning (mean over held-out cities). Disp.: mean distance between true and predicted cell edges; length: relative error of the number of cell-edge pixels. Plan: sites chosen greedily on the predicted map until it covers 90% of the pixels at 140 dB, and the true coverage of those sites; truth: sites the same procedure needs on the true map.

| Predictor                       | Cell edges |            | Plan for 90% |               |
|---------------------------------|------------|------------|--------------|---------------|
|                                 | disp. (m)  | length (%) | sites        | true cov. (%) |
| Tile labels (oracle)            | 76         | -92        | 1.1          | 49.7          |
| Source median                   | 80         | -93        | 1.0          | 56.4          |
| 3GPP UMa NLoS                   | 80         | -93        | 5.6          | 84.0          |
| Kernel (NW)                     | 80         | -93        | 1.0          | 51.9          |
| GP-ARD                          | 80         | -93        | 1.0          | 50.7          |
| Ridge                           | 80         | -93        | 1.0          | 50.8          |
| GBDT                            | 80         | -93        | 1.0          | 47.4          |
| GBDT + 10 tiles                 | 80         | -93        | 1.0          | 49.3          |
| Link transfer (ours)            | 7          | -5         | 3.8          | 81.6          |
| Link, regressor only            | 7          | -4         | 1.0          | 56.9          |
| Link + 10 designed tiles (ours) | 7          | -3         | 3.9          | 81.9          |
| Truth                           | 0          | 0          | 6.7          | 90+           |

Table 10: Link transfer: the ten features with the largest share of the total split gain (mean over folds and seeds).

| Feature                  | gain share (%) |
|--------------------------|----------------|
| knife-edge loss          | 22.9           |
| max. excess height       | 19.7           |
| buildings crossed        | 17.4           |
| log distance             | 11.7           |
| blocked fraction         | 8.5            |
| length over buildings    | 5.9            |
| horizontal distance      | 2.4            |
| built fraction, 50 m     | 1.8            |
| receiver building height | 1.8            |
| site height              | 1.6            |

*Network size.* On the network of the first 10 sites (Table C.20, Appendix C), the ranking is unchanged: the oracle has a path-loss error of 11.7 dB, GBDT tile transfer 13.0 dB and link transfer 9.2 dB, with serving-site accuracies of 51.3%, 51.3% and 68.3%.

*Siting policy.* On the 4 cities re-traced with sites on random rooftops (median site height 26 m; Table 14), the link models trained on the tallest-rooftop corpus reach a path-loss error of 10.9 dB, against 12.0 dB for the tile model with the exact labels of the new network, and a planning regret of 1.9% against 15.7%. The tile labels of the same tiles fitted under the original siting give 15.9 dB: the labels of a tile do not carry over to other sites, as the dependence of the exponent on the serving sites (Section 5.2) implies.

*Second band.* At 7.5 GHz the corpus has 7,038,989 links below the cut-off (10,274 labelled tiles). The tile exponents are lower than at 3.5 GHz (median 2.18 against 2.79) and the intercepts higher by a median of 7.9 dB, against 6.6 dB for free space; the per-tile exponents of the two bands correlate with  $r = 0.74$ – $0.92$  across cities. The lower exponents agree with the stronger censoring of the higher band. The ranking of the predictors is

Table 11: Receiver-tile and mixed-path compositions of the tile model, and link transfer, on the held-out cities (mean over cities). Tile labels: fitted to the links of the held-out city itself; median, kernel and GBDT: transferred from the training cities. PL: path-loss RMSE; serv.: serving-site accuracy; SINR: mean absolute error; edge: cell-edge displacement; regret:  $K = 5$ .

| Predictor                                  | PL (dB) | serv. (%) | SINR (dB) | edge (m) | regret (%) |
|--------------------------------------------|---------|-----------|-----------|----------|------------|
| <i>Receiver tile (Proposition 1)</i>       |         |           |           |          |            |
| Tile labels (oracle)                       | 11.64   | 50.1      | 7.56      | 76       | 9.3        |
| GBDT                                       | 12.80   | 50.1      | 7.44      | 80       | 20.6       |
| <i>Mixed path, path-averaged</i>           |         |           |           |          |            |
| Tile labels (oracle)                       | 11.14   | 54.1      | 6.97      | 25       | 5.3        |
| Source median                              | 14.27   | 50.1      | 7.76      | 80       | 31.3       |
| Kernel (NW)                                | 13.72   | 44.8      | 7.51      | 50       | 21.7       |
| GBDT                                       | 13.45   | 46.3      | 7.66      | 49       | 18.3       |
| <i>Mixed path, piecewise (multi-slope)</i> |         |           |           |          |            |
| Tile labels (oracle)                       | 11.63   | 49.1      | 7.06      | 29       | 14.3       |
| Source median                              | 14.20   | 50.3      | 7.62      | 79       | 30.4       |
| Kernel (NW)                                | 15.85   | 37.9      | 7.63      | 69       | 27.4       |
| GBDT                                       | 16.94   | 34.3      | 7.69      | 59       | 24.1       |
| <i>Mixed path, summed as in [8]</i>        |         |           |           |          |            |
| Tile labels (oracle)                       | 33.20   | 21.3      | 11.15     | 14       | 7.9        |
| Source median                              | 92.66   | 48.7      | 7.07      | 54       | 30.9       |
| Kernel (NW)                                | 74.49   | 46.5      | 8.31      | 39       | 27.9       |
| GBDT                                       | 75.14   | 41.4      | 9.61      | 35       | 29.2       |
| <i>Site-aware link model</i>               |         |           |           |          |            |
| Link transfer (ours)                       | 9.33    | 68.8      | 5.37      | 7        | 0.5        |

Table 12: Per-site calibration of the link model from  $k$  surveyed tiles: random tiles (mean of ten draws) and the survey design of Proposition 5. PL: path-loss RMSE; serv.: serving-site accuracy; regret:  $K = 5$ . Means over the held-out cities. Zero-shot link model: 9.33 dB, 68.8%, 0.5%.

| $k$ | Random survey |           |            | Designed survey |           |            |
|-----|---------------|-----------|------------|-----------------|-----------|------------|
|     | PL (dB)       | serv. (%) | regret (%) | PL (dB)         | serv. (%) | regret (%) |
| 10  | 9.46          | 65.9      | 0.5        | 9.52            | 65.6      | 0.9        |
| 25  | 9.35          | 67.6      | 0.5        | 9.46            | 66.0      | 0.7        |
| 50  | 9.30          | 69.1      | 0.5        | 9.44            | 67.0      | 0.5        |

unchanged (Table 15): the tile labels give 10.9 dB, GBDT tile transfer 12.0 dB and link transfer trained at 7.5 GHz 8.5 dB (better than GBDT in 11 of 11 cities). The link models trained at 3.5 GHz transfer across bands with the free-space offset alone: 9.0 dB, and 9.1 dB with the knife-edge feature evaluated at the new wavelength, both below the tile model with the exact labels of the new band, with a serving-site accuracy of 68.9% and a planning regret of 1.0%.

*External scenes.* On the 3 independently modelled scenes (Table 16; 202,782 links), whose geometry and materials differ from our corpus, zero-shot link transfer remains better than transferred tile parameters in path loss (13.1 dB against 14.9 dB for GBDT, in 3 of 3 scenes), serving site (68.5% against 55.4%), SINR (5.0 dB against 5.3 dB) and cell edges (9 m against 45 m), but its path-loss error is above that of the tile model with the exact labels of these scenes (12.24 dB) and it has a positive bias of 2.9 to 4.8 dB, consistent with the different materials of these scenes (brick, marble, metal and wood, against the single concrete model of the training corpus). This is the systematic shift that the per-site calibration of Section 4.6 is for: with 25 randomly surveyed tiles the bias falls to 0.0 dB on average and the path-loss error to 12.26 dB, equal to that of the exact tile labels (lower in 1 of 3 scenes), while the serving-

Table 13: Certified coverage (Proposition 6), mean over held-out cities.  $\hat{q}_t$ : margin from the other cities; valid: cities in which  $q_t \leq \hat{q}_t$  (of 11);  $r_p \leq \hat{q}_t$ : share of pixels within the margin; certified claim: share claimed covered and share claimed but not covered; naive claim: without a margin; covered: true covered share.

| Predictor                                                                                  | $\hat{q}_t$ (dB) | valid | $r_p \leq \hat{q}_t$ (%) | Certified claim |           | Naive claim |           | covered (%) |
|--------------------------------------------------------------------------------------------|------------------|-------|--------------------------|-----------------|-----------|-------------|-----------|-------------|
|                                                                                            |                  |       |                          | claimed (%)     | false (%) | claimed (%) | false (%) |             |
| $\alpha = 0.1, L = 140$ dB, <i>bound holding with probability <math>\geq 10/11</math></i>  |                  |       |                          |                 |           |             |           |             |
| Tile labels (oracle)                                                                       | 13.4             | 10    | 93.9                     | 88.2            | 1.79      | 99.9        | 3.62      | 96.3        |
| Source median                                                                              | 22.2             | 10    | 95.5                     | 60.5            | 1.60      | 100.0       | 3.68      | 96.3        |
| 3GPP UMa NLoS                                                                              | 14.6             | 10    | 95.3                     | 55.6            | 1.32      | 95.5        | 3.48      | 96.3        |
| Kernel (NW)                                                                                | 19.3             | 10    | 95.7                     | 72.3            | 1.44      | 100.0       | 3.68      | 96.3        |
| Ridge                                                                                      | 17.8             | 10    | 95.1                     | 77.6            | 1.60      | 100.0       | 3.68      | 96.3        |
| GBDT                                                                                       | 17.6             | 10    | 95.1                     | 78.3            | 1.66      | 100.0       | 3.68      | 96.3        |
| Link transfer (ours)                                                                       | 9.8              | 10    | 93.9                     | 92.0            | 1.79      | 98.3        | 3.11      | 96.3        |
| Link, regressor only                                                                       | 10.3             | 10    | 94.0                     | 91.3            | 1.69      | 100.0       | 3.68      | 96.3        |
| $\alpha = 0.1, L = 140$ dB, <i>bound holding with probability <math>\geq 9/11</math></i>   |                  |       |                          |                 |           |             |           |             |
| Tile labels (oracle)                                                                       | 13.2             | 9     | 93.7                     | 88.9            | 1.88      | 99.9        | 3.62      | 96.3        |
| Source median                                                                              | 22.1             | 9     | 95.4                     | 61.0            | 1.64      | 100.0       | 3.68      | 96.3        |
| 3GPP UMa NLoS                                                                              | 14.3             | 9     | 95.1                     | 56.5            | 1.36      | 95.5        | 3.48      | 96.3        |
| Kernel (NW)                                                                                | 18.1             | 9     | 94.8                     | 77.2            | 1.80      | 100.0       | 3.68      | 96.3        |
| Ridge                                                                                      | 15.2             | 9     | 93.4                     | 86.7            | 2.26      | 100.0       | 3.68      | 96.3        |
| GBDT                                                                                       | 16.3             | 9     | 94.2                     | 83.2            | 2.00      | 100.0       | 3.68      | 96.3        |
| Link transfer (ours)                                                                       | 8.9              | 9     | 93.1                     | 93.5            | 2.03      | 98.3        | 3.11      | 96.3        |
| Link, regressor only                                                                       | 9.8              | 9     | 93.4                     | 92.4            | 1.88      | 100.0       | 3.68      | 96.3        |
| $\alpha = 0.05, L = 140$ dB, <i>bound holding with probability <math>\geq 10/11</math></i> |                  |       |                          |                 |           |             |           |             |
| Tile labels (oracle)                                                                       | 18.7             | 10    | 97.3                     | 71.0            | 0.75      | 99.9        | 3.62      | 96.3        |
| Source median                                                                              | 27.5             | 10    | 98.0                     | 32.4            | 0.64      | 100.0       | 3.68      | 96.3        |
| 3GPP UMa NLoS                                                                              | 20.4             | 10    | 97.9                     | 34.2            | 0.63      | 95.5        | 3.48      | 96.3        |
| Kernel (NW)                                                                                | 24.5             | 10    | 98.1                     | 48.8            | 0.50      | 100.0       | 3.68      | 96.3        |
| Ridge                                                                                      | 22.5             | 10    | 97.7                     | 57.8            | 0.64      | 100.0       | 3.68      | 96.3        |
| GBDT                                                                                       | 22.3             | 10    | 97.6                     | 58.8            | 0.68      | 100.0       | 3.68      | 96.3        |
| Link transfer (ours)                                                                       | 14.5             | 10    | 97.1                     | 82.7            | 0.82      | 98.3        | 3.11      | 96.3        |
| Link, regressor only                                                                       | 14.7             | 10    | 97.1                     | 82.3            | 0.78      | 100.0       | 3.68      | 96.3        |

site accuracy rises to 71.9% against 55.4% for the tile labels. The designed survey of 25 tiles reaches only 13.16 dB. With 8 sites on these smaller scenes, five sites cover almost every pixel, and the zero-shot predictors all have the same planning regret.

### 5.11. Statistical summary

Tables D.22 and D.23 (Appendix D) list all 71 city-level comparisons. Link transfer improves on GBDT tile transfer in path loss, serving site, coverage agreement and planning regret in all 11 cities, and on the tile model with the true labels in path loss, serving site and planning regret in all 11 cities; these comparisons remain significant after Holm’s correction. At the parameter level the evidence is weaker: of the 12 comparisons of a transfer operator with the source median, no improvement survives Holm’s correction at eleven cities, whereas the gain of GP-ARD over kernel transfer does.

## 6. Discussion

*The tile exponent is not an environmental constant.* The central premise of descriptor-based tile transfer is that a tile has a path-loss exponent determined by its environment, which can be read from descriptors and carried to similar tiles elsewhere. On this corpus the premise is weak, for three measured reasons. Two

estimates of the same tile’s exponent from disjoint halves of the sites differ by 1.37 in standard deviation, so the exponent depends on which sites serve the tile; it depends on the dynamic range of the measurements, shifting by  $-0.583$  when the cut-off is lowered by 10 dB; and the cross-city spread of the city medians (0.27) is small against the spread within a city. A tile is seen by each site through a different street canyon, and the exponent fitted over all of them mixes the geometry of the tile with that of the paths reaching it. This explains why learned operators improve on the source median only modestly, why few-shot correction of the exponent needs tens of surveyed tiles before it helps, and why a planning tool fed with even the *true* receiver-tile labels assigns half of the pixels to the wrong site and draws cell edges 76 m away from the true ones.

*What to transfer.* The quantity that transfers is the effect of the geometry along a link, not a per-tile summary of it. Site-aware link transfer uses the same descriptors, the same training cities and the same protocol, but conditions on the path between site and receiver; it improves path loss, serving site, coverage and planning in every held-out city and outperforms the tile model with perfect labels, however the tile parameters are composed along the path and whatever the tile size (Section 5.7). It keeps its advantage over transferred tile parameters at another frequency, with other sites and on independently modelled scenes.

Table 14: Siting policy: models of the main corpus applied to 4 cities (Amsterdam, London, New York, Paris) re-traced with sites on random rooftops. Left: the same predictors on the tallest-rooftop networks of these cities. Tile labels of this network: fitted on the random-site links; main siting: the labels of the same tiles in the main corpus. Means over cities (and seeds).

| Predictor                   | Tallest rooftops |           |            | Random rooftops |           |           |            |
|-----------------------------|------------------|-----------|------------|-----------------|-----------|-----------|------------|
|                             | PL (dB)          | serv. (%) | regret (%) | PL (dB)         | serv. (%) | SINR (dB) | regret (%) |
| Tile labels of this network | 11.71            | 50.1      | 8.0        | 12.04           | 54.3      | 12.53     | 15.7       |
| Tile labels, main siting    | –                | –         | –          | 15.87           | 54.8      | 12.21     | 19.1       |
| Source median               | 14.76            | 50.1      | 34.5       | 15.24           | 54.9      | 11.68     | 42.8       |
| GBDT                        | 12.98            | 50.1      | 16.5       | 14.79           | 54.9      | 11.78     | 29.9       |
| Link transfer (ours)        | 9.35             | 66.8      | 1.0        | 10.85           | 65.6      | 9.52      | 1.9        |

Table 15: Second band (7.5 GHz), mean over held-out cities. Above the rule: models trained at 7.5 GHz; below: the link models trained at 3.5 GHz with the free-space offset, with the knife-edge feature at the new wavelength or unchanged.

| Predictor                        | PL (dB) | serv. (%) | SINR (dB) | regret (%) |
|----------------------------------|---------|-----------|-----------|------------|
| Tile labels (oracle)             | 10.92   | 49.6      | 7.87      | 7.6        |
| Source median                    | 13.25   | 49.8      | 7.54      | 37.5       |
| 3GPP UMa NLoS                    | 24.88   | 49.8      | 7.27      | 8.6        |
| GBDT                             | 11.95   | 49.8      | 7.67      | 22.5       |
| Link transfer (ours)             | 8.52    | 67.6      | 5.67      | 0.9        |
| Link, 3.5 GHz model (ours)       | 9.08    | 68.9      | 5.33      | 1.0        |
| Link, 3.5 GHz model, offset only | 8.99    | 68.9      | 5.30      | 0.9        |

On the latter, whose materials differ from the training corpus, it is biased until a survey of 25 random tiles calibrates it, after which its path loss matches exact tile labels and its serving-site accuracy exceeds theirs: the material models of the training corpus limit what transfers zero-shot, and a small unbiased survey corrects the rest. Its most important features, the knife-edge loss and the height of the obstruction above the line of sight, are the quantities of classical diffraction theory [50], so the model remains interpretable in physical terms. Two design choices matter for planning. First, the model must represent the absence of a link: the hurdle, which predicts whether a site is received at all, reduces the planning regret from 5.6% to 0.5% at the same path-loss error. Second, its errors must be stated at the level of the decision: the city-level certificate turns the leave-one-city-out residuals into a coverage promise with a guaranteed risk, and the more accurate the model, the more of the city it can promise (92.0% against 78.3% for tile transfer). For a digital twin this means that the transferable representation should be attached to links (or to site–tile pairs), which is also what a planning tool evaluates.

*Kernel transfer.* Theorem 1 identifies two limits of kernel transfer; the tuned operator is in neither, but with the bandwidths selected on two targets it approaches the source mean without reaching the flat regime. Its disadvantage against the learned regressors is the noise-transmission term of Proposition 2 combined with an isotropic distance: learning one length scale per descriptor (GP-ARD) removes most of the gap, which locates the weakness in the metric rather than in kernel averaging as such.

*Implications for planning.* Four practical conclusions follow. First, a transfer model for a digital twin should be validated at the network level; parameter errors of similar size lead to very different planning losses (Table 8), and a uniformly biased model such as the 3GPP UMa NLoS constants has a path-loss error of 22.8 dB, far above all transfer operators, yet a planning regret of 8.0%, below that of transferred tile parameters, because a uniform bias does not change which sites cover the most pixels. Second, models trained on links below a receiver cut-off overstate the reach of distant sites unless the cut-off is modelled; every zero-shot tile model with transferred parameters predicts that a single site covers 90% of a city. Third, small surveys do not help a model without systematic error, but correct a shifted one, and they must be unbiased: a design that maximises the variance reduction of the offsets selects unrepresentative tiles. Fourth, coverage claims should come with a margin estimated across cities: without it, 3.68% of the pixels promised by tile transfer are not covered.

*Limitations.* The path loss is ray traced rather than measured, so absolute values depend on the solver, the ITU material models and the level-of-detail-1 geometry; the results are comparisons under identical conditions. The external scenes test a different geometry source and different materials but the same solver, and the siting replicate and the second band test other sites and another frequency; none replaces validation against measurements, which the public drive-test data we could obtain do not allow because they lack site positions. Ray tracing is nevertheless a demanding rather than a lenient testbed for tile transfer: its losses arise site by site from the building geometry of each path, not from a statistical model with tile structure, so a tile model has no advantage built in, and it provides what measurements cannot, a complete map for every site and exact labels for every tile, which separate the error of the tile model from that of transfer. Between 2% and 50% of the building heights are imputed per city. With eleven cities, city-level tests have limited power: an exact Wilcoxon test with Holm’s correction over a dozen comparisons requires nearly every city to agree, which is why no improvement of a tile operator over the median survives it although the learned operators improve on it in eight or nine of eleven cities. The corpus has flat terrain, 20 rooftop sites per city with isotropic antennas, full load and a fixed link budget. Tile labels come from the same ray tracer as the network truth, so the oracle is the reference for tile models

Table 16: External scenes of Sionna RT (no retraining). PL: path-loss RMSE; bias: mean path-loss error, predicted minus true (dB); serv.: serving-site accuracy; SINR: mean absolute error; edge: cell-edge displacement; regret:  $K = 5$ . Tile labels: fitted on the scene itself; median, ridge and GBDT: fitted on the eleven cities; link: mean of the eleven leave-one-city-out link models, zero-shot and calibrated from 25 surveyed tiles (random: mean of ten draws; designed).

| Predictor                                          | PL (dB) | bias | serv. (%) | SINR (dB) | edge (m) | regret (%) |
|----------------------------------------------------|---------|------|-----------|-----------|----------|------------|
| <i>Munich</i> (104,696 links, 8 sites, 195 tiles)  |         |      |           |           |          |            |
| Tile labels (oracle)                               | 12.09   | +0.0 | 63.4      | 5.55      | 48       | 1.6        |
| Source median                                      | 15.33   | +7.1 | 63.4      | 4.56      | 48       | 1.6        |
| Ridge                                              | 14.66   | +3.4 | 63.4      | 4.54      | 48       | 1.6        |
| GBDT                                               | 14.48   | +2.8 | 63.4      | 4.56      | 48       | 1.6        |
| Link transfer (ours)                               | 13.29   | +4.2 | 67.8      | 4.46      | 10       | 1.6        |
| Link + 25 tiles (ours)                             | 12.41   | +0.5 | 65.2      | 4.59      | 10       | 0.1        |
| Link + 25 designed tiles                           | 14.03   | +6.2 | 56.7      | 4.77      | 11       | 0.1        |
| <i>Étoile</i> (42,235 links, 8 sites, 63 tiles)    |         |      |           |           |          |            |
| Tile labels (oracle)                               | 10.69   | +0.0 | 55.1      | 4.44      | 38       | 2.0        |
| Source median                                      | 15.12   | +8.5 | 55.1      | 4.67      | 38       | 2.0        |
| Ridge                                              | 13.91   | +2.5 | 55.1      | 4.83      | 38       | 2.0        |
| GBDT                                               | 13.75   | +2.1 | 55.1      | 4.83      | 38       | 2.0        |
| Link transfer (ours)                               | 11.96   | +4.8 | 74.9      | 4.40      | 6        | 2.0        |
| Link + 25 tiles (ours)                             | 10.96   | -0.5 | 84.1      | 2.65      | 4        | 0.1        |
| Link + 25 designed tiles                           | 11.34   | -3.0 | 86.9      | 2.71      | 3        | 0.0        |
| <i>Florence</i> (55,851 links, 8 sites, 127 tiles) |         |      |           |           |          |            |
| Tile labels (oracle)                               | 13.93   | +0.0 | 47.6      | 7.37      | 44       | 2.2        |
| Source median                                      | 15.51   | +1.1 | 47.6      | 6.52      | 48       | 2.2        |
| Ridge                                              | 16.72   | +0.1 | 47.6      | 6.69      | 48       | 2.2        |
| GBDT                                               | 16.45   | -0.9 | 47.6      | 6.62      | 48       | 2.2        |
| Link transfer (ours)                               | 14.12   | +2.9 | 62.6      | 6.02      | 10       | 2.2        |
| Link + 25 tiles (ours)                             | 13.42   | +0.0 | 66.2      | 5.53      | 10       | 0.0        |
| Link + 25 designed tiles                           | 14.11   | +4.7 | 65.2      | 5.41      | 11       | 0.0        |
| <i>Mean over scenes</i>                            |         |      |           |           |          |            |
| Tile labels (oracle)                               | 12.24   | +0.0 | 55.4      | 5.79      | 43       | 1.9        |
| Source median                                      | 15.32   | +5.6 | 55.4      | 5.25      | 45       | 1.9        |
| Ridge                                              | 15.10   | +2.0 | 55.4      | 5.35      | 45       | 1.9        |
| GBDT                                               | 14.89   | +1.3 | 55.4      | 5.34      | 45       | 1.9        |
| Link transfer (ours)                               | 13.12   | +4.0 | 68.5      | 4.96      | 9        | 1.9        |
| Link + 25 tiles (ours)                             | 12.26   | +0.0 | 71.9      | 4.26      | 8        | 0.1        |
| Link + 25 designed tiles                           | 13.16   | +2.6 | 69.6      | 4.30      | 8        | 0.1        |

with exact labels on this corpus, not a statement about measured data.

## 7. Conclusion

We built reproducible ray-traced corpora of 11 cities in two bands for cross-city transfer of path-loss models and evaluated transfer where a network planner uses it. Tile labels of the log-distance law are ill-conditioned and, once their correlated errors are accounted for, imprecise: the cluster-robust standard error of the exponent is 3.2 times the naive one, and the exponent depends on the serving sites, the tile size and the cut-off of the data. Transferring tile parameters with learned regressors improves on the source median in 9 of 11 cities, although not significantly after correction for multiple comparisons; kernel transfer improves on it less, and our analysis of its limiting regimes and the gain from learning its metric locate its weakness in the isotropic distance. At the network level, even perfect tile parameters leave a path-loss error of 11.6 dB, assign only 50.1% of the pixels to the correct site and misplace cell

edges by 76 m; composing the parameters of the tiles along each path improves on this with exact labels, but not in path loss or serving site once they are transferred. Site-aware hurdle link transfer, which predicts each link and whether it exists from its obstruction geometry, reduces the path-loss error to 9.3 dB, raises serving-site accuracy to 68.8% and lowers the planning regret in every held-out city, and a city-level certificate turns its accuracy into coverage that can be promised at a stated risk.

Future work should validate these findings on measured data with known site positions, extend the corpora to terrain, antenna patterns and further bands, and study certificates that adapt to the target city from a small survey.

## CRedit authorship contribution statement

**Abraheem Rashid:** Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft. **Faisal Iradat:** Software, Validation, Investigation. **Waseem Iqbal:** Supervision, Writing – review & editing. **Yawar Abbas Bangash:** Formal analysis, Writing – review &

editing. **Bilgin Metin:** Supervision, Writing – review & editing.

## Declaration of competing interest

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During the preparation of this work the authors used Claude (Anthropic) to assist with software development, data analysis scripting and the drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## Data availability

The code, the scene definitions, the ray-traced path-loss corpus, the tile tables and every result file are available at <https://github.com/Abraheem13/terra-net>. Running make reproduce regenerates all numbers, tables and figures of this article and the article itself. Building footprints are from Overture Maps (release 2026-09-23.0), © OpenStreetMap contributors, Overture Maps Foundation, under the Open Database License.

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## Appendix A. Proofs

*Proposition 1.* With the parameters of tile  $t$  for every site,  $P_{pb} = P_{tx} - P_{L_{0,t}} - 10\gamma_t \log_{10}(d_{pb}/d_0)$  is strictly decreasing in  $d_{pb}$  when  $\gamma_t > 0$ , which gives (i). Writing  $10^{P_{pb}/10} = C(d_{pb}/d_0)^{-\gamma_t}$  with  $C = 10^{(P_{tx} - P_{L_{0,t}})/10}$  in Eq. (1) and dividing numerator and denominator by  $C$  gives Eq. (3).  $\square$

*Proposition 2.* With  $\hat{\theta}(\mathbf{v}) = \sum_j w_j(\mathbf{v})(\theta^*(\mathbf{v}_j) + \eta_j)$  and  $\sum_j w_j = 1$ ,  $\hat{\theta} - \theta^*(\mathbf{v}) = [\sum_j w_j \theta^*(\mathbf{v}_j) - \theta^*(\mathbf{v})] + \sum_j w_j \eta_j$ . The weights depend on descriptors only and  $\mathbb{E}[\eta_j] = \mathbf{0}$  independently of the

descriptors, so the cross term has zero expectation, and independence across  $j$  gives  $\mathbb{E}[\|\sum_j w_j \eta_j\|^2] = \text{tr}(\Sigma_\eta) \sum_j w_j^2$ .  $\square$

*Proposition 3.*  $\mathbf{A}^\top \mathbf{A} = [\sum_i x_i^2 \quad \sum_i x_i; \sum_i x_i \quad n]$  has determinant  $n \sum x_i^2 - (\sum x_i)^2 = n^2 \text{Var}(x)$ , and the (1,1) entry of its inverse is  $n/(n^2 \text{Var}(x))$ , so  $\text{Var}(\hat{\gamma}_{\text{OLS}}) = \sigma_\epsilon^2/(n \text{Var}(x))$ .  $\square$

*Proposition 4.* Given the true slope,  $b_t$  has mean  $\gamma_t$  and variance  $v_t$ ; the true slopes have mean  $\mu$  and variance  $\tau^2$ . For fixed  $w$ ,  $\hat{\gamma}_t - \gamma_t = (1 - w)(b_t - \gamma_t) + w(\mu - \gamma_t)$ ; the two terms are uncorrelated, so the mean squared error is  $(1 - w)^2 v_t + w^2 \tau^2$ , a convex quadratic in  $w$  minimised at  $w_t = v_t/(v_t + \tau^2)$  with value  $v_t \tau^2/(v_t + \tau^2)$ , which is at most  $\min(v_t, \tau^2)$ . With centred  $x$ , the ridge problem  $\min_\gamma \sum_i (y_i - \bar{y} - \gamma x_i)^2 + \lambda(\gamma - \mu)^2$  has solution  $(S_{xy} + \lambda\mu)/(S_{xx} + \lambda) = (1 - w)b_t + w\mu$  with  $w = \lambda/(\lambda + S_{xx})$ ; setting  $\lambda_t = S_{xx}v_t/\tau^2$  gives  $w = v_t/(v_t + \tau^2) = w_t$ .  $\square$

*Theorem 1, part (a).* Put  $\alpha = 1/(2\sigma^2)$ . The factor  $e^{-\alpha \min_k d_k}$  cancels in Eq. (7), so  $\tilde{w}_j = e^{-\alpha \delta_j} / \sum_k e^{-\alpha \delta_k}$  with  $\alpha \delta_j \leq \epsilon_u < 1$ . By Taylor’s theorem,  $e^{-\alpha \delta_j} = 1 - \alpha \delta_j + r_j$  with  $0 \leq r_j \leq \epsilon_u^2/2$ , hence  $\sum_k e^{-\alpha \delta_k} = N(1 - \alpha \bar{\delta}) + O(N\epsilon_u^2)$  and

$$\tilde{w}_j = \frac{1}{N}(1 - \alpha(\delta_j - \bar{\delta})) + O(\epsilon_u^2/N).$$

Then  $\hat{\theta}_u = \sum_j \tilde{w}_j \theta_j = \bar{\theta} - \alpha \text{Cov}_j(\delta_j, \theta_j) + O(\epsilon_u^2)$ , where the remainder is bounded by a constant times  $\epsilon_u^2 \max_j \|\theta_j - \bar{\theta}\|$ . By Cauchy–Schwarz,  $\|\text{Cov}_j(\delta_j, \theta_j)\| \leq \sigma_\delta \sigma_\theta$ , and by Popoviciu’s inequality  $\sigma_\delta \leq D_u/2$ , which gives the bound in Eq. (8). Squaring the weight expansion,  $\sum_j \tilde{w}_j^2 = N^{-1}(1 + \alpha^2 \sigma_\delta^2) + O(\epsilon_u^2/N)$ , the linear term vanishing because  $\sum_j (\delta_j - \bar{\delta}) = 0$ ; since  $\alpha \sigma_\delta \leq \epsilon_u/2$ ,  $n_{\text{eff}} = N(1 + O(\epsilon_u^2))$ .  $\square$

*Theorem 1, part (b).* Order the source tiles by distance, so  $\delta_{(1)} = 0$  and  $\delta_{(j)} \geq g_u$  for  $j \geq 2$ . Then  $\tilde{w}_{(1)} = (1 + \sum_{j \geq 2} e^{-\alpha \delta_{(j)}})^{-1} \geq (1 + (N - 1)e^{-\alpha g_u})^{-1}$ , and  $\|\hat{\theta}_u - \theta_{(1)}\| = \|\sum_{j \geq 2} \tilde{w}_{(j)}(\theta_{(j)} - \theta_{(1)})\| \leq (1 - \tilde{w}_{(1)}) \max_j \|\theta_j - \theta_{(1)}\| \leq (N - 1)e^{-\alpha g_u} \max_j \|\theta_j - \theta_{(1)}\|$ .  $\square$

*Proposition 5.*  $n_s(T)$  is a non-negative modular function of  $T$  and  $\phi(n) = n/(n + \kappa)$  is non-decreasing and concave on  $[0, \infty)$ . For  $A \subseteq B$  and a tile  $t \notin B$ ,  $n_s(A) \leq n_s(B)$  and concavity give  $\phi(n_s(A) + c_{ts}) - \phi(n_s(A)) \geq \phi(n_s(B) + c_{ts}) - \phi(n_s(B)) \geq 0$ . Summing over  $s$  shows that  $F$  is monotone and has diminishing returns, i.e. is submodular, and  $F(\emptyset) = 0$ ; the greedy bound is the theorem of [28]. The posterior variance follows from the conjugate normal model: the precision of  $o_s$  is  $1/\tau_o^2 + n_s/\sigma_\epsilon^2$ , so its variance is  $\tau_o^2 \kappa/(n_s + \kappa)$  and the reduction is  $\tau_o^2 n_s/(n_s + \kappa)$ .  $\square$

*Proposition 6.* Each  $q_c$  is a function of city  $c$  and of the multiset of the other cities, through the same symmetric procedure, so exchangeable cities give an exchangeable vector  $(q_1, \dots, q_n)$ . The event that  $q_t$  exceeds the  $j$ -th largest of the other values means that  $q_t$  is among the  $j$  largest of the vector, with no tie at the  $j$ -th place; by exchangeability its probability is at most  $j/n$ , hence  $\Pr(q_t \leq \hat{q}_t) \geq (n - j)/n$ . On this event, a claimed pixel with  $r_p \leq \hat{q}_t$  satisfies  $PL_{\text{serv}}(p) = \bar{PL}_{\text{serv}}(p) + r_p \leq \bar{PL}_{\text{serv}}(p) + \hat{q}_t \leq L$ , so it is covered. Claimed but uncovered pixels therefore have  $r_p > \hat{q}_t \geq q_t$ , and by the definition of the quantile at most a fraction  $\alpha$  of the pixels do.  $\square$

## Appendix B. Selected hyper-parameters

Table B.17: Hyper-parameters selected on the validation city of each split.  $N$ : training tiles;  $\sigma$ : kernel bandwidth;  $k$ : neighbours; trees: median number of boosting rounds for  $\gamma$  over the five seeds.

| Target    | Validation | $N$   | $\sigma$ | $k$  | $\alpha_{\text{ridge}}$ | $\alpha_{\text{MLP}}$ | trees |
|-----------|------------|-------|----------|------|-------------------------|-----------------------|-------|
| Amsterdam | Prague     | 8,520 | 0.1      | 50   | 0.01                    | 0.01                  | 105   |
| Barcelona | New York   | 8,526 | 0.5      | 1000 | 100                     | 1                     | 9     |
| Berlin    | Munich     | 8,521 | 0.1      | 200  | 0.01                    | 0.01                  | 80    |
| Chicago   | Berlin     | 8,510 | 0.1      | 200  | 0.01                    | 0.01                  | 48    |
| Hamburg   | Chicago    | 8,506 | 0.5      | 1000 | 100                     | 1                     | 8     |
| London    | Amsterdam  | 8,540 | 0.2      | 50   | 0.01                    | 0.01                  | 102   |
| Munich    | Amsterdam  | 8,530 | 0.2      | 50   | 0.01                    | 0.01                  | 118   |
| New York  | Amsterdam  | 8,520 | 0.1      | 50   | 0.01                    | 0.0001                | 96    |
| Paris     | Barcelona  | 8,576 | 0.1      | 20   | 0.01                    | 0.01                  | 97    |
| Prague    | Paris      | 8,558 | 0.1      | 200  | 0.01                    | 0.01                  | 85    |
| Vienna    | Munich     | 8,539 | 0.1      | 200  | 0.01                    | 0.01                  | 57    |

## Appendix C. Additional robustness results

Tables C.18–C.20 give the full results of the tile-size, cut-off and network-size studies summarised in Section 5.10, and Table C.21 the mixed-path compositions of Section 5.7 at the tile sizes of [8].

Table C.18: Tile size. Tiles per city, median shrinkage weight (mean of the city medians) and standard error of the exponent (iid and cluster-robust, RMS over all tiles); zero-shot exponent RMSE of the source median and GBDT; network path-loss RMSE of the tile labels (oracle) and of GBDT tile transfer; serving-site accuracy of the tile model. Means over the held-out cities.

| $G_s$ (m) | tiles/city | med. $w_t$ | se( $\gamma$ ) |         | $\gamma$ RMSE |       | PL RMSE (dB) |       | serv. (%) |
|-----------|------------|------------|----------------|---------|---------------|-------|--------------|-------|-----------|
|           |            |            | iid            | cluster | median        | GBDT  | oracle       | GBDT  |           |
| 50        | 3470       | 0.38       | 0.234          | 0.540   | 0.985         | 0.888 | 11.19        | 12.67 | 50.1      |
| 100       | 948        | 0.37       | 0.141          | 0.456   | 0.871         | 0.782 | 11.64        | 12.80 | 50.1      |
| 200       | 244        | 0.38       | 0.088          | 0.369   | 0.716         | 0.619 | 12.09        | 13.05 | 50.1      |

Table C.19: Path-loss cut-off. Share of (pixel, site) pairs of the labelled tiles beyond the 150 dB cut-off and of links above 140 dB; median change of the tile exponent when the cut-off is lowered to 140 and 130 dB; for random tiles, median and 90th percentile of the Tobit minus the least-squares slope.

| City      | censored<br>pairs (%) | Lower cut-off   |                      |                      | Tobit vs least squares |                     |        |
|-----------|-----------------------|-----------------|----------------------|----------------------|------------------------|---------------------|--------|
|           |                       | links > 140 (%) | $\Delta\gamma_{140}$ | $\Delta\gamma_{130}$ | tiles                  | med. $\Delta\gamma$ | q90    |
| Amsterdam | 56.8                  | 18.4            | −0.572               | −1.141               | 300                    | +3.746              | +6.802 |
| Barcelona | 61.0                  | 18.7            | −0.486               | −0.893               | 300                    | +4.284              | +7.509 |
| Berlin    | 43.5                  | 17.3            | −0.646               | −1.261               | 300                    | +3.256              | +6.938 |
| Chicago   | 40.5                  | 17.9            | −0.640               | −1.226               | 300                    | +3.536              | +6.698 |
| Hamburg   | 35.5                  | 15.0            | −0.504               | −1.066               | 300                    | +2.113              | +3.886 |
| London    | 46.9                  | 14.6            | −0.470               | −0.855               | 300                    | +2.946              | +5.014 |
| Munich    | 54.7                  | 20.1            | −0.583               | −1.184               | 300                    | +3.157              | +5.333 |
| New York  | 53.8                  | 23.8            | −0.621               | −1.119               | 300                    | +4.492              | +7.343 |
| Paris     | 63.1                  | 18.3            | −0.598               | −1.006               | 300                    | +4.266              | +8.338 |
| Prague    | 55.8                  | 21.5            | −0.729               | −1.366               | 300                    | +3.500              | +5.315 |
| Vienna    | 66.4                  | 22.9            | −0.527               | −0.980               | 300                    | +4.146              | +7.915 |

Table C.20: Network size: the same predictors on the 20-site and on the 10-site network (mean over held-out cities). PL: path-loss RMSE; serv.: serving-site accuracy; SINR: mean absolute error; regret:  $K = 5$ .

| Predictor            | 20 sites |           |           |            | 10 sites |           |           |            |
|----------------------|----------|-----------|-----------|------------|----------|-----------|-----------|------------|
|                      | PL (dB)  | serv. (%) | SINR (dB) | regret (%) | PL (dB)  | serv. (%) | SINR (dB) | regret (%) |
| Tile labels (oracle) | 11.64    | 50.1      | 7.56      | 9.3        | 11.67    | 51.3      | 8.05      | 6.3        |
| Source median        | 14.52    | 50.1      | 7.18      | 32.1       | 14.91    | 51.3      | 7.85      | 27.0       |
| 3GPP UMa NLoS        | 22.75    | 50.1      | 6.92      | 8.0        | 23.87    | 51.3      | 8.04      | 4.5        |
| Kernel (NW)          | 13.02    | 50.1      | 7.47      | 19.0       | 13.20    | 51.3      | 8.07      | 18.7       |
| GP-ARD               | 12.80    | 50.1      | 7.56      | 17.9       | 12.95    | 51.3      | 8.12      | 12.7       |
| Ridge                | 12.80    | 50.1      | 7.54      | 18.9       | 12.96    | 51.3      | 8.11      | 14.7       |
| GBDT                 | 12.80    | 50.1      | 7.44      | 20.6       | 12.95    | 51.3      | 8.04      | 16.1       |
| Link transfer (ours) | 9.33     | 68.8      | 5.37      | 0.5        | 9.22     | 68.3      | 6.22      | 0.4        |

Table C.21: Mixed-path compositions of the tile model for tile sizes  $G_s$  of 100 m (the corpus) and 200 m, 500 m and 1000 m (the range of [8]), on the pixels of the main corpus (mean over held-out cities). Tiles: tiles per city crossed by at least 30 links; labels: parameters fitted to the held-out city itself; GBDT: transferred from the training cities. PL: path-loss RMSE (dB); serv.: serving-site accuracy (%). Link transfer: 9.3 dB, 68.8%.

| $G_s$ (m) |     | Summed as in [8] |       |      |       | Piecewise |       |      |       | Path-averaged |       |      |       |
|-----------|-----|------------------|-------|------|-------|-----------|-------|------|-------|---------------|-------|------|-------|
|           |     | labels           |       | GBDT |       | labels    |       | GBDT |       | labels        |       | GBDT |       |
|           |     | PL               | serv. | PL   | serv. | PL        | serv. | PL   | serv. | PL            | serv. | PL   | serv. |
| 100       | 951 | 33.2             | 21.3  | 75.1 | 41.4  | 11.6      | 49.1  | 16.9 | 34.3  | 11.1          | 54.1  | 13.4 | 46.3  |
| 200       | 241 | 40.0             | 29.7  | 61.2 | 41.7  | 12.2      | 49.3  | 15.2 | 40.7  | 11.8          | 51.5  | 13.5 | 46.8  |
| 500       | 49  | 42.6             | 33.7  | 49.5 | 35.8  | 12.7      | 47.4  | 14.7 | 38.1  | 12.5          | 47.7  | 14.0 | 47.5  |
| 1000      | 14  | 42.7             | 37.7  | 48.3 | 44.5  | 13.1      | 47.0  | 16.1 | 44.6  | 13.0          | 48.5  | 14.8 | 48.5  |

## Appendix D. Complete city-level comparisons

Tables D.22 and D.23 list every city-level comparison behind the statistical statements of Section 5.11.

Table D.22: City-level comparisons of the tile parameters ( $n = 11$  held-out cities).  $\Delta$ : mean difference first minus second; CI: 95% percentile bootstrap over cities;  $p$ : exact two-sided Wilcoxon signed-rank; better: cities in which the first is better; Holm: significant at 5% after Holm’s correction within the family.

| Comparison                                          | Metric        | $\Delta$ | 95% CI           | $p$    | better | Holm |
|-----------------------------------------------------|---------------|----------|------------------|--------|--------|------|
| <i>Zero-shot vs source median</i>                   |               |          |                  |        |        |      |
| CORAL + ridge vs Source median                      | $\gamma$ RMSE | -0.085   | [-0.143, -0.026] | 0.024  | 8/11   | no   |
| COST-231 Hata vs Source median                      | $\gamma$ RMSE | +0.183   | [+0.093, +0.269] | 0.005  | 1/11   | no   |
| Neural encoder vs Source median                     | $\gamma$ RMSE | -0.105   | [-0.158, -0.047] | 0.010  | 9/11   | no   |
| GBDT vs Source median                               | $\gamma$ RMSE | -0.089   | [-0.132, -0.043] | 0.010  | 9/11   | no   |
| GBDT-IW vs Source median                            | $\gamma$ RMSE | -0.088   | [-0.135, -0.034] | 0.014  | 9/11   | no   |
| GP-ARD vs Source median                             | $\gamma$ RMSE | -0.107   | [-0.160, -0.049] | 0.010  | 9/11   | no   |
| Kernel (NW) vs Source median                        | $\gamma$ RMSE | -0.059   | [-0.111, +0.005] | 0.147  | 9/11   | no   |
| $k$ -NN vs Source median                            | $\gamma$ RMSE | -0.088   | [-0.148, -0.022] | 0.024  | 8/11   | no   |
| MLP vs Source median                                | $\gamma$ RMSE | -0.110   | [-0.166, -0.049] | 0.010  | 9/11   | no   |
| Ridge vs Source median                              | $\gamma$ RMSE | -0.103   | [-0.158, -0.042] | 0.019  | 9/11   | no   |
| 3GPP UMa LoS vs Source median                       | $\gamma$ RMSE | +0.131   | [+0.054, +0.205] | 0.042  | 4/11   | no   |
| 3GPP UMa NLoS vs Source median                      | $\gamma$ RMSE | +0.581   | [+0.462, +0.698] | <0.001 | 0/11   | yes  |
| <i>Zero-shot pairs</i>                              |               |          |                  |        |        |      |
| GBDT vs Kernel (NW)                                 | $\gamma$ RMSE | -0.030   | [-0.057, -0.008] | 0.054  | 9/11   | no   |
| GBDT vs Ridge                                       | $\gamma$ RMSE | +0.014   | [-0.010, +0.042] | 0.831  | 6/11   | no   |
| Kernel (NW) vs $k$ -NN                              | $\gamma$ RMSE | +0.030   | [-0.002, +0.062] | 0.147  | 3/11   | no   |
| GP-ARD vs Kernel (NW)                               | $\gamma$ RMSE | -0.049   | [-0.075, -0.025] | 0.005  | 9/11   | yes  |
| GBDT-IW vs GBDT                                     | $\gamma$ RMSE | +0.001   | [-0.016, +0.016] | 0.278  | 4/11   | no   |
| CORAL + ridge vs Ridge                              | $\gamma$ RMSE | +0.018   | [-0.004, +0.043] | 0.320  | 5/11   | no   |
| <i>Few-shot correction</i>                          |               |          |                  |        |        |      |
| GBDT, $k=10$ vs GBDT, $k=0$                         | $\gamma$ RMSE | +0.003   | [-0.018, +0.021] | 0.365  | 3/11   | no   |
| GBDT, $k=10$ designed vs GBDT, $k=10$               | $\gamma$ RMSE | -0.007   | [-0.022, +0.010] | 0.240  | 7/11   | no   |
| Kernel (NW), $k=10$ vs Kernel (NW), $k=0$           | $\gamma$ RMSE | +0.000   | [-0.012, +0.011] | 1.000  | 5/11   | no   |
| Kernel (NW), $k=10$ designed vs Kernel (NW), $k=10$ | $\gamma$ RMSE | -0.002   | [-0.024, +0.028] | 0.413  | 7/11   | no   |
| MLP, $k=10$ vs MLP, $k=0$                           | $\gamma$ RMSE | +0.018   | [+0.009, +0.026] | 0.010  | 1/11   | no   |
| MLP, $k=10$ designed vs MLP, $k=10$                 | $\gamma$ RMSE | -0.008   | [-0.028, +0.018] | 0.365  | 8/11   | no   |
| Ridge, $k=10$ vs Ridge, $k=0$                       | $\gamma$ RMSE | +0.013   | [+0.004, +0.020] | 0.024  | 1/11   | no   |
| Ridge, $k=10$ designed vs Ridge, $k=10$             | $\gamma$ RMSE | -0.010   | [-0.027, +0.011] | 0.278  | 8/11   | no   |

Table D.23: City-level comparisons at the network level, as in Table D.22 ( $\Delta$  in percentage points for shares: serving site, coverage agreement, regret).

| Comparison                                                | Metric              | $\Delta$ | 95% CI         | $p$    | better | Holm |
|-----------------------------------------------------------|---------------------|----------|----------------|--------|--------|------|
| <i>Network level</i>                                      |                     |          |                |        |        |      |
| GBDT vs Source median                                     | path-loss RMSE (dB) | -1.72    | [-2.26, -1.21] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs GBDT                              | path-loss RMSE (dB) | -3.47    | [-3.90, -3.12] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | path-loss RMSE (dB) | -2.32    | [-2.61, -2.07] | <0.001 | 11/11  | yes  |
| Link + 10 tiles (ours) vs Link transfer (ours)            | path-loss RMSE (dB) | +0.13    | [+0.11, +0.16] | <0.001 | 0/11   | yes  |
| Link + 10 designed tiles (ours) vs Link + 10 tiles (ours) | path-loss RMSE (dB) | +0.06    | [-0.01, +0.15] | 0.278  | 5/11   | no   |
| Link + 50 designed tiles (ours) vs Link transfer (ours)   | path-loss RMSE (dB) | +0.12    | [+0.05, +0.19] | 0.007  | 1/11   | no   |
| GBDT vs Source median                                     | serving site        | +0.0     | [+0.0, +0.0]   | 1.000  | 0/11   | no   |
| Link transfer (ours) vs GBDT                              | serving site        | +18.7    | [+15.1, +22.6] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | serving site        | +18.7    | [+15.3, +22.7] | <0.001 | 11/11  | yes  |
| Link + 10 tiles (ours) vs Link transfer (ours)            | serving site        | -2.9     | [-4.5, -1.7]   | <0.001 | 0/11   | yes  |
| Link + 10 designed tiles (ours) vs Link + 10 tiles (ours) | serving site        | -0.2     | [-1.4, +1.0]   | 0.765  | 6/11   | no   |
| Link + 50 designed tiles (ours) vs Link transfer (ours)   | serving site        | -1.8     | [-4.1, -0.2]   | 0.067  | 4/11   | no   |
| GBDT vs Source median                                     | SINR MAE (dB)       | +0.26    | [+0.16, +0.37] | <0.001 | 0/11   | yes  |
| Link transfer (ours) vs GBDT                              | SINR MAE (dB)       | -2.07    | [-2.45, -1.75] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | SINR MAE (dB)       | -2.20    | [-2.52, -1.91] | <0.001 | 11/11  | yes  |
| Link + 10 tiles (ours) vs Link transfer (ours)            | SINR MAE (dB)       | +0.15    | [+0.10, +0.21] | <0.001 | 0/11   | yes  |
| Link + 10 designed tiles (ours) vs Link + 10 tiles (ours) | SINR MAE (dB)       | +0.14    | [+0.06, +0.22] | 0.014  | 3/11   | no   |
| Link + 50 designed tiles (ours) vs Link transfer (ours)   | SINR MAE (dB)       | +0.12    | [+0.00, +0.27] | 0.240  | 5/11   | no   |
| GBDT vs Source median                                     | coverage agreement  | +0.4     | [-0.4, +1.5]   | 0.898  | 5/11   | no   |
| Link transfer (ours) vs GBDT                              | coverage agreement  | +2.1     | [+1.3, +2.9]   | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | coverage agreement  | +0.8     | [+0.3, +1.2]   | 0.002  | 10/11  | yes  |
| Link + 10 tiles (ours) vs Link transfer (ours)            | coverage agreement  | -0.1     | [-0.2, +0.1]   | 0.413  | 4/11   | no   |
| Link + 10 designed tiles (ours) vs Link + 10 tiles (ours) | coverage agreement  | -0.2     | [-0.4, -0.1]   | 0.014  | 2/11   | no   |
| Link + 50 designed tiles (ours) vs Link transfer (ours)   | coverage agreement  | -0.3     | [-0.5, -0.1]   | 0.054  | 2/11   | no   |
| GBDT vs Source median                                     | regret, $K=5$       | -11.5    | [-19.1, -3.8]  | 0.019  | 10/11  | no   |
| Link transfer (ours) vs GBDT                              | regret, $K=5$       | -20.1    | [-27.0, -14.1] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | regret, $K=5$       | -8.8     | [-11.6, -6.1]  | <0.001 | 11/11  | yes  |
| Link + 10 tiles (ours) vs Link transfer (ours)            | regret, $K=5$       | +0.0     | [+0.0, +0.1]   | 0.312  | 4/11   | no   |
| Link + 10 designed tiles (ours) vs Link + 10 tiles (ours) | regret, $K=5$       | +0.4     | [+0.0, +1.1]   | 0.109  | 1/11   | no   |
| Link + 50 designed tiles (ours) vs Link transfer (ours)   | regret, $K=5$       | +0.0     | [+0.0, +0.1]   | 0.250  | 0/11   | no   |
| <i>Network level, second band</i>                         |                     |          |                |        |        |      |
| Link transfer (ours) vs GBDT                              | path-loss RMSE (dB) | -3.43    | [-3.85, -3.10] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | path-loss RMSE (dB) | -2.40    | [-2.69, -2.16] | <0.001 | 11/11  | yes  |
| Link, 3.5 GHz model (ours) vs Link transfer (ours)        | path-loss RMSE (dB) | +0.56    | [+0.42, +0.70] | <0.001 | 0/11   | yes  |
| Link transfer (ours) vs GBDT                              | serving site        | +17.8    | [+14.6, +21.5] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | serving site        | +17.9    | [+14.8, +21.6] | <0.001 | 11/11  | yes  |
| Link, 3.5 GHz model (ours) vs Link transfer (ours)        | serving site        | +1.4     | [+1.0, +1.7]   | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs GBDT                              | SINR MAE (dB)       | -2.00    | [-2.28, -1.73] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | SINR MAE (dB)       | -2.20    | [-2.48, -1.96] | <0.001 | 11/11  | yes  |
| Link, 3.5 GHz model (ours) vs Link transfer (ours)        | SINR MAE (dB)       | -0.34    | [-0.46, -0.23] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs GBDT                              | coverage agreement  | +6.8     | [+4.8, +9.1]   | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | coverage agreement  | +5.3     | [+3.7, +7.2]   | <0.001 | 11/11  | yes  |
| Link, 3.5 GHz model (ours) vs Link transfer (ours)        | coverage agreement  | -3.7     | [-4.9, -2.8]   | <0.001 | 0/11   | yes  |
| Link transfer (ours) vs GBDT                              | regret, $K=5$       | -21.7    | [-28.5, -15.9] | <0.001 | 11/11  | yes  |
| Link transfer (ours) vs Tile labels (oracle)              | regret, $K=5$       | -6.7     | [-9.5, -4.3]   | <0.001 | 11/11  | yes  |
| Link, 3.5 GHz model (ours) vs Link transfer (ours)        | regret, $K=5$       | +0.2     | [-0.2, +0.5]   | 0.461  | 4/11   | no   |